



sAlfer Lab

Joint lab on Safety and Security of AI

Computer Vision Technologies and Biometrics

Multimodal biometrics

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**M.Sc. Degree in Computer Engineering, CyberSecurity and
Artificial Intelligence**



University
of Cagliari, Italy

Department of Electrical
and Electronic
Engineering



Outline of the talk

- **Behavioural Biometrics and Behaviour Recognition**
- **Face expressions recognition**
- **Gestural analysis**
- **Detection of anomalous events in small and large crowds**

From behavioural biometrics to behaviour recognition



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<https://www.youtube.com/watch?v=-SZMtaSO8Ow>

Face expression recognition



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Sistema riconoscimento espressioni facciali

Enter the image path:



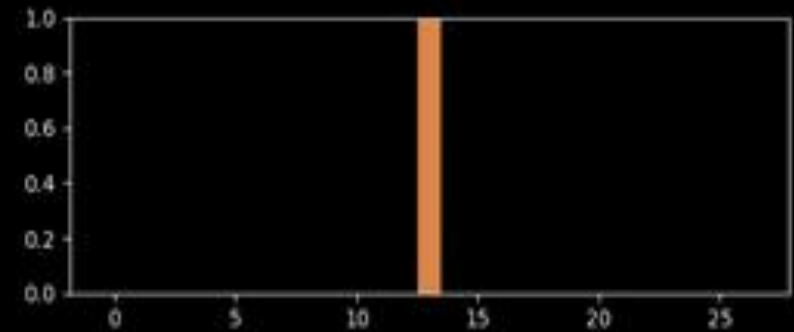
Analyze Expression

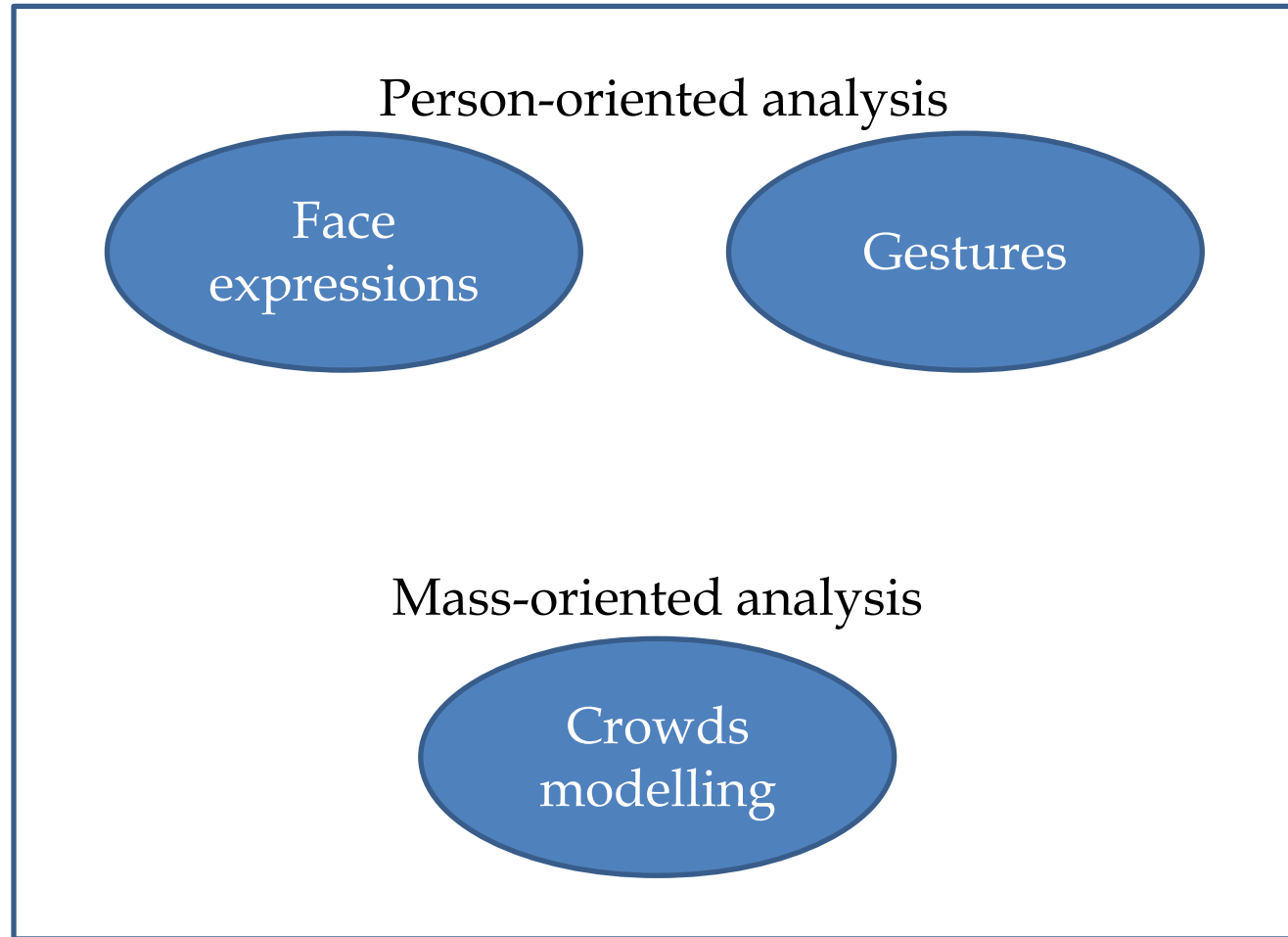
Expression:

Anomalous events in crowds

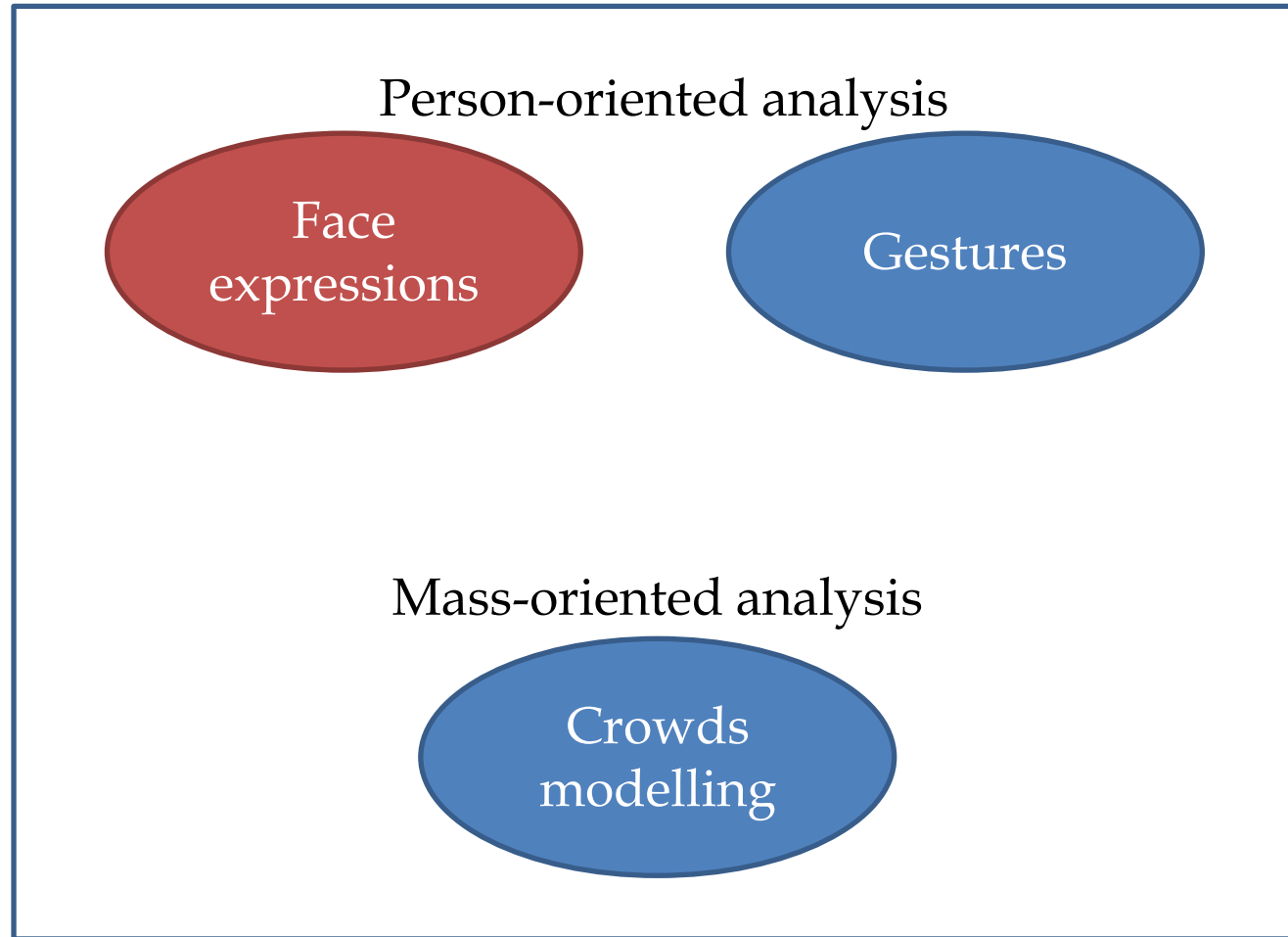


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**Emotion and behaviour
recognition**

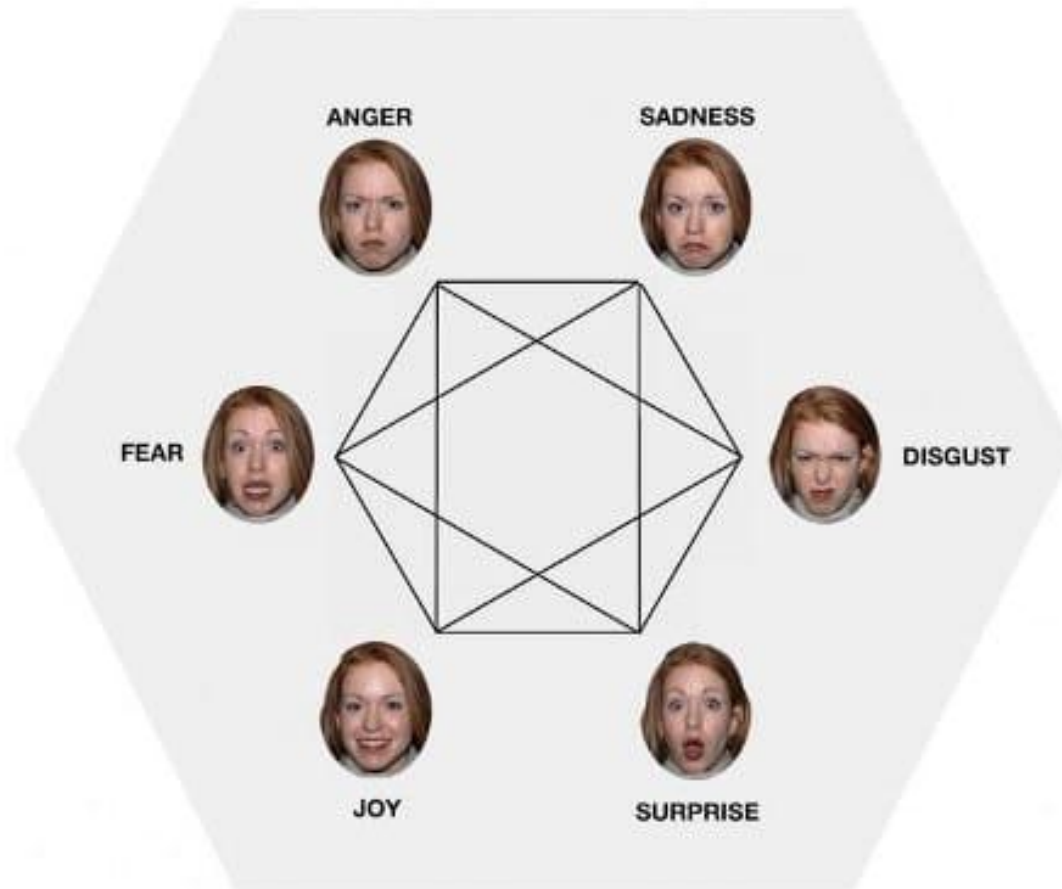


**Emotion and behaviour
recognition**

Facial expressions and emotions

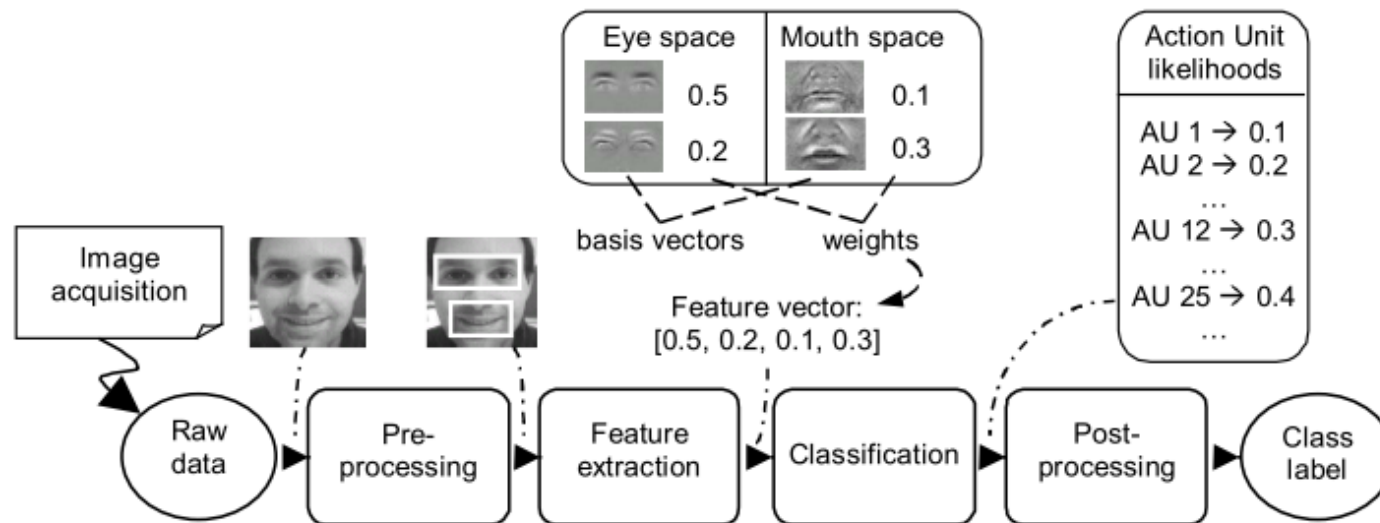


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P. Ekman (1978) and W. V. Friesen (2002)
Facial Action Coding System (FACS)

Overall traditional scheme



Example	2D image showing a front or profile view of the face [15]	Geometric normalisation against head translation, rotation, scale [6] [12]. Face segmentation using a Gaussian mixture model of the skin [19], or deformable models of face parts [15] [19].	Estimation of geometric measurements between facial landmarks [1]. Space dimensionality reduction for dense motion parameters [12]. Estimation of muscle activation parameters [9].	<u>Classifier</u> Hidden Markov models [12]. Neural network [19]. k-nearest neighbour classifier [1]. Rule-based expert system [15].	Multi-stage classification [15]	Joy
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Facial Action Units



Lower Face Action Units					
AU 9	AU 10	AU 11	AU 12	AU 13	AU 14
					
Nose Wrinkler	Upper Lip Raiser	Nasolabial Deepener	Lip Corner Puller	Cheek Puffer	Dimpler
AU 15	AU 16	AU 17	AU 18	AU 20	AU 22
					
Lip Corner Depressor	Lower Lip Depressor	Chin Raiser	Lip Pucker	Lip Stretcher	Lip Funneler
AU 23	AU 24	*AU 25	*AU 26	*AU 27	AU 28
					
Lip Tightener	Lip Pressor	Lips Part	Jaw Drop	Mouth Stretch	Lip Suck

Facial Action Units



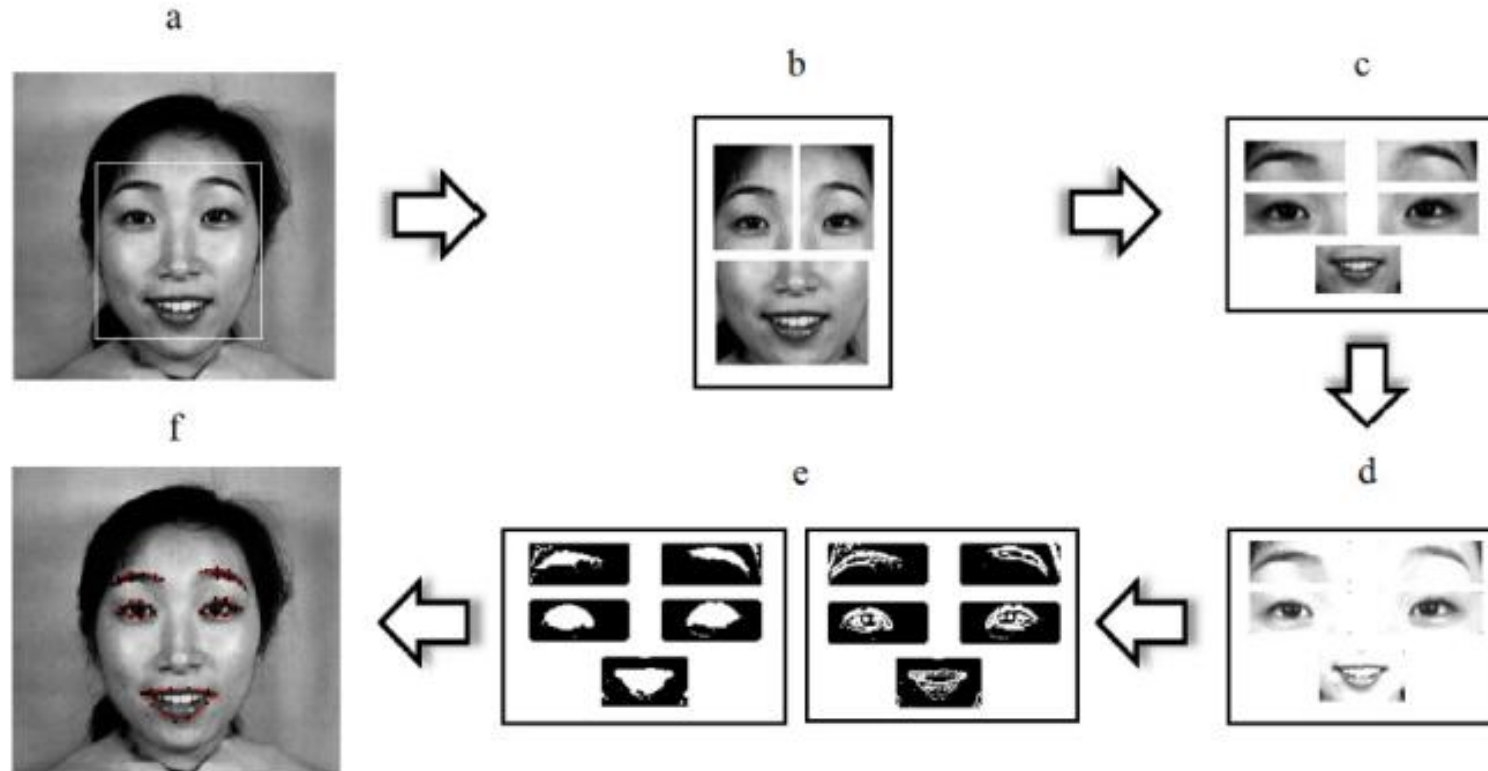
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Upper Face Action Units					
AU1	AU2	AU4	AU5	AU6	AU7
					
Inner Brow Raiser	Outer Brow Raiser	Brow Lowerer	Upper Lid Raiser	Cheek Raiser	Lid Tightener
*AU41	*AU42	*AU43	AU44	AU45	AU46
					
Lip Droop	Slit	Eyes Closed	Squint	Blink	Wink
Lower Face Action Units					
AU9	AU10	AU11	AU12	AU13	AU14
					
Nose Wrinkler	Upper Lip Raiser	Nasolabial Deepener	Lip Corner Puller	Cheek Puffer	Dimpler
AU15	AU16	AU17	AU18	AU20	AU22
					
Lip Corner Depressor	Lower Lip Depressor	Chin Raiser	Lip Puckerer	Lip Stretcher	Lip Funneler
AU23	AU24	*AU25	*AU26	*AU27	AU28
					
Lip Tightener	Lip Pressor	Lips Parts	Jaw Drop	Mouth Stretch	Lip Suck

FAC by CV



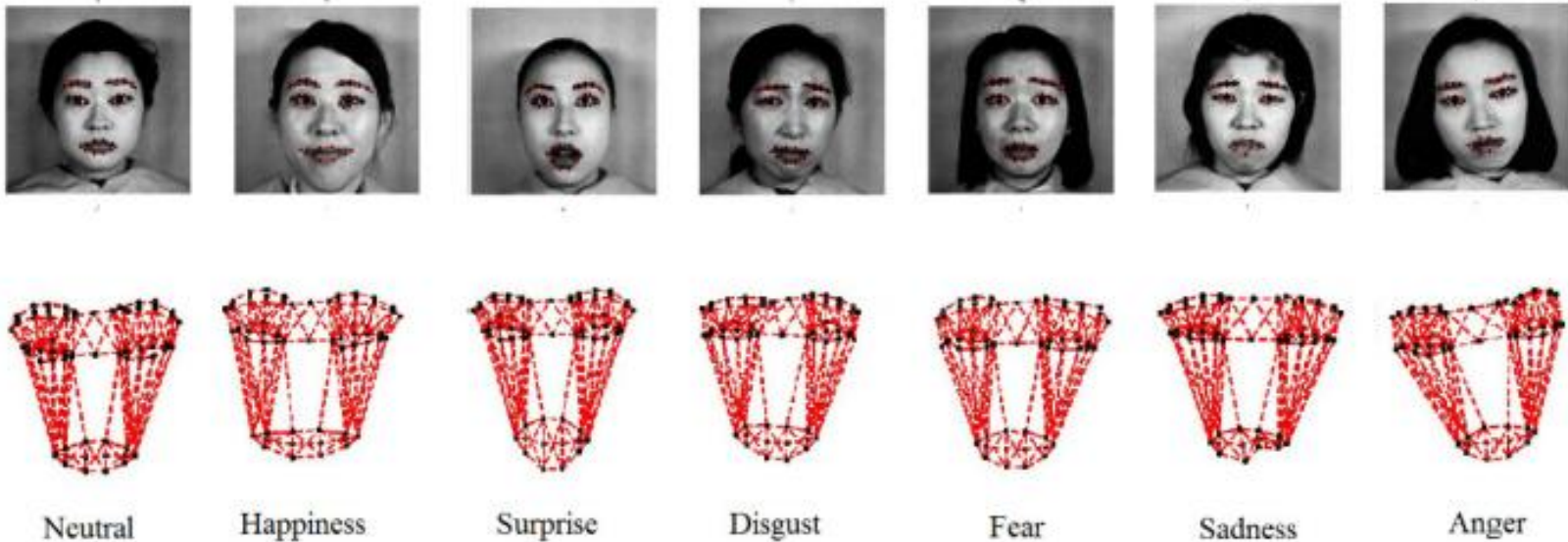
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Graph-based representation



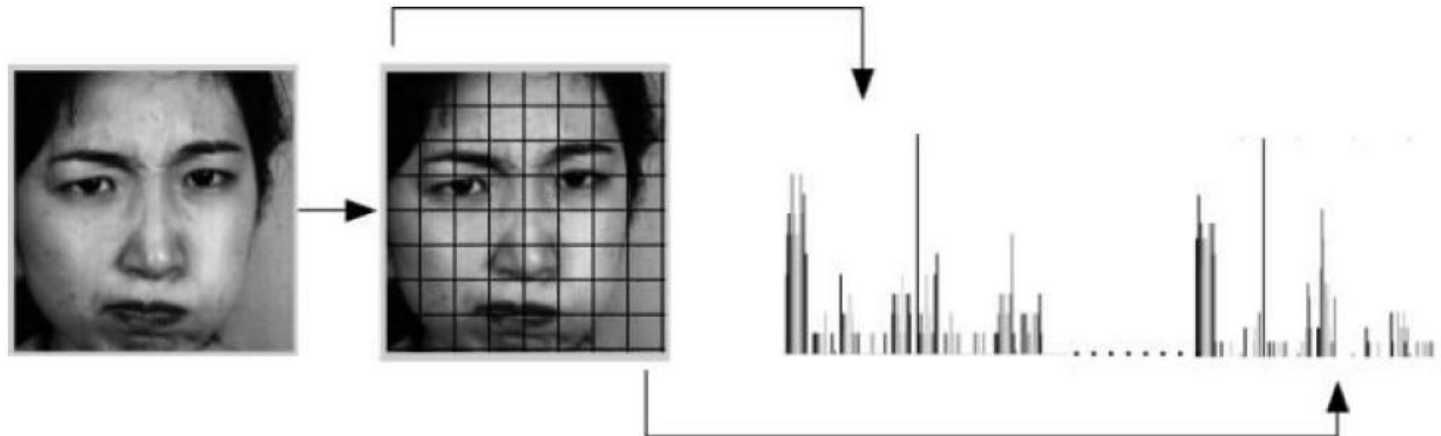
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➤ Over-time structure

- Elastic Bunch Graphs (semi-independent from AUs)
- Action Units based Graph (46)

Texture-based approaches



- **Local Binary Patterns (LBP)**
- **Local Gradient Code (LGC)**
- **Local Directional Patterns (LDP)**
- **Histogram of Gradient Orientation (HOG)**

Local Gradient Code



g₁	g₂	g₃
g₄	g_c	g₅
g₆	g₇	g₈

$$LGC = s(g_1 - g_3)2^7 + s(g_4 - g_5)2^6 + s(g_6 - g_8)2^5 + s(g_1 - g_6)2^4 + s(g_2 - g_7)2^3 + s(g_3 - g_8)2^2 + s(g_1 - g_8)2^1 + s(g_3 - g_6)2^0$$

$$LGC - HD_p^R = s(g_1 - g_3)2^4 + s(g_4 - g_5)2^3 + s(g_6 - g_8)2^2 + s(g_1 - g_8)2^1 + s(g_3 - g_6)2^0$$

$$LGC - VD_p^R = s(g_1 - g_6)2^4 + s(g_2 - g_7)2^3 + s(g_3 - g_8)2^2 + s(g_1 - g_8)2^1 + s(g_3 - g_6)2^0$$

LGC «template»

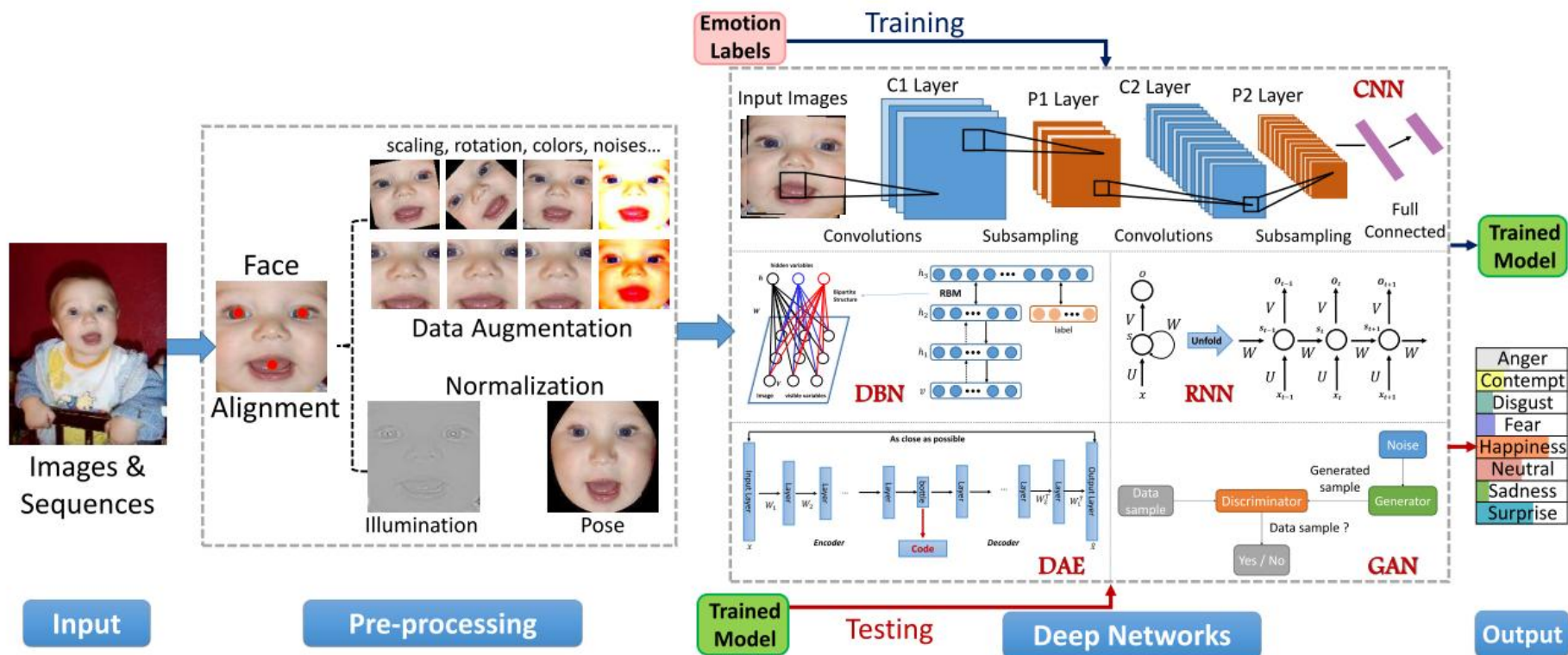
Table 1. Results of FER on JAFFE dataset for 256 x 256 image sizes with 16 x 16 blocks.

Methods	Feature Length	Recognition Rate
LBP	65536	89.4231
LGC	65536	90.3846
LGC-HD	65536	87.50
LGC-VD	65536	92.3077
HOG	20736	85.7143
LDP	14337	85.2041

Deep learning methods: the pipeline



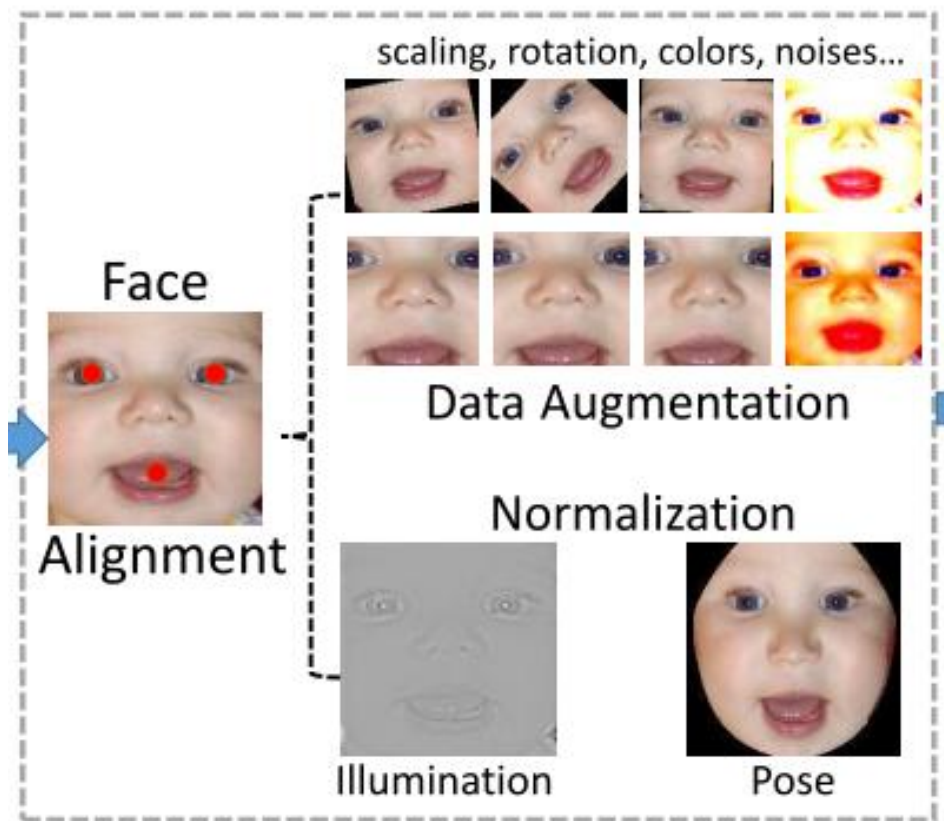
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Pre-processing

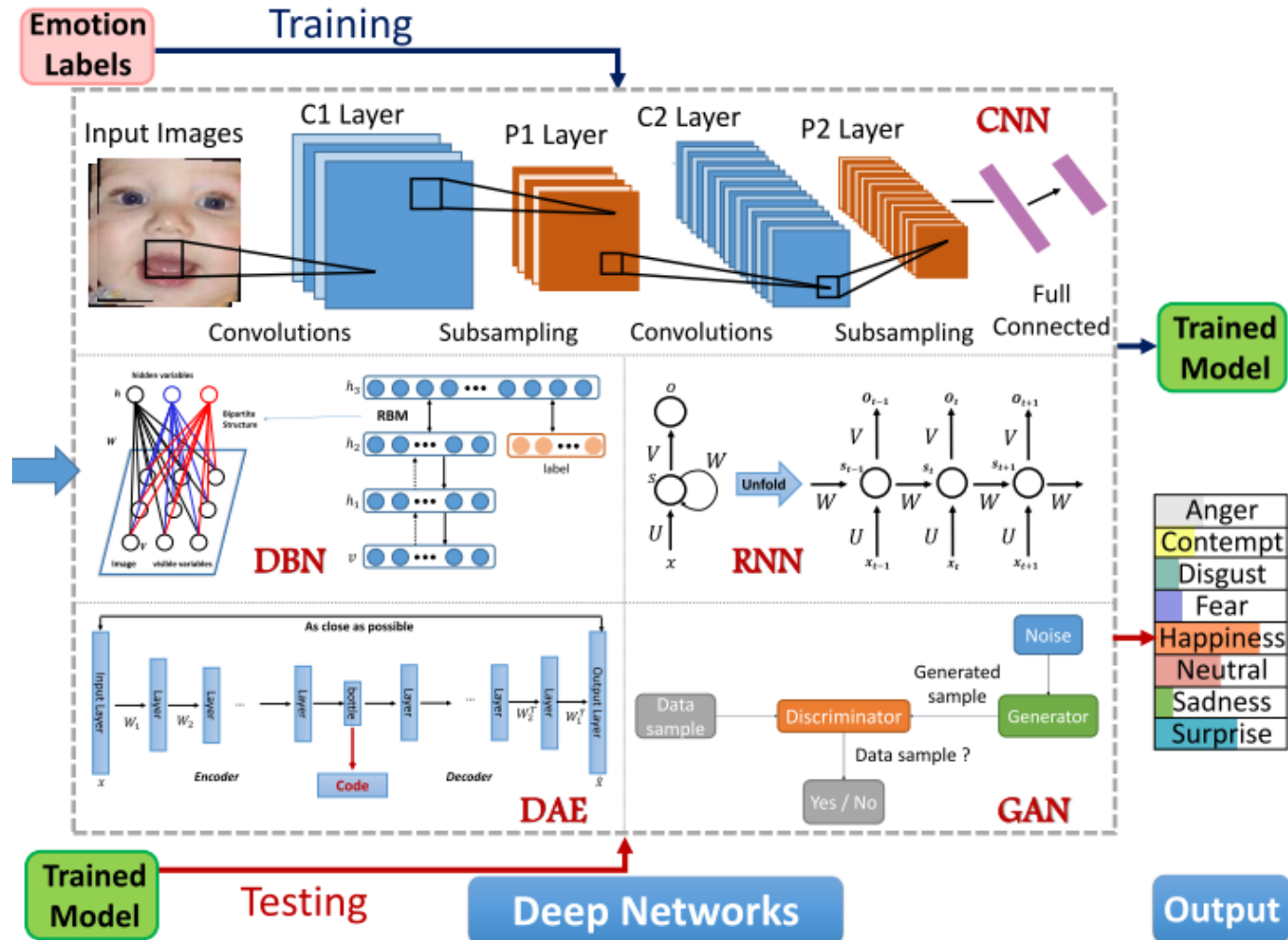


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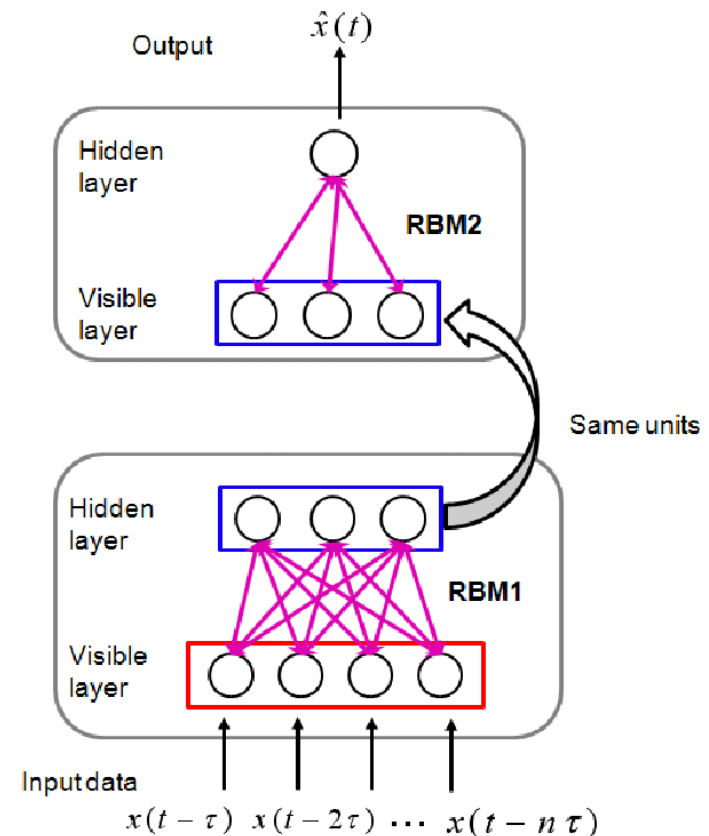
Pre-processing

Deep networks





- **Convolutional Neural Networks (CNN)**
 - Based on «convolutional» layers whose coefficient are learnt as well as the weights of the fully connected part
 - They work as a «first-level» feature extractors from the pattern
- **Deep Belief Networks (DBN)**
 - First group of neurons is binary probability layers (Restricted Boltzmann Machine)
 - Second group sets the weights of the probabilities approximated by the first one
 - Characteristic: one layer is trained at a time
 - Training is done in unsupervised manner

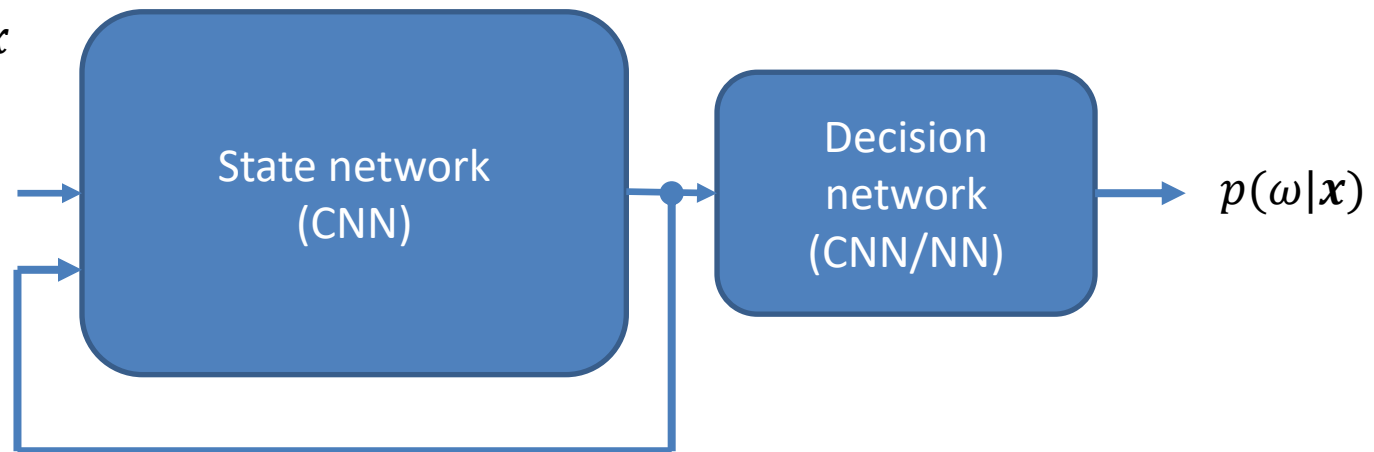




- **Recursive/Recurrent Neural Networks (RNN)**

- See the example about fingerprint classification
- Recurrent NNs take as input temporal sequences
- Recursive NNs take as input structured patterns

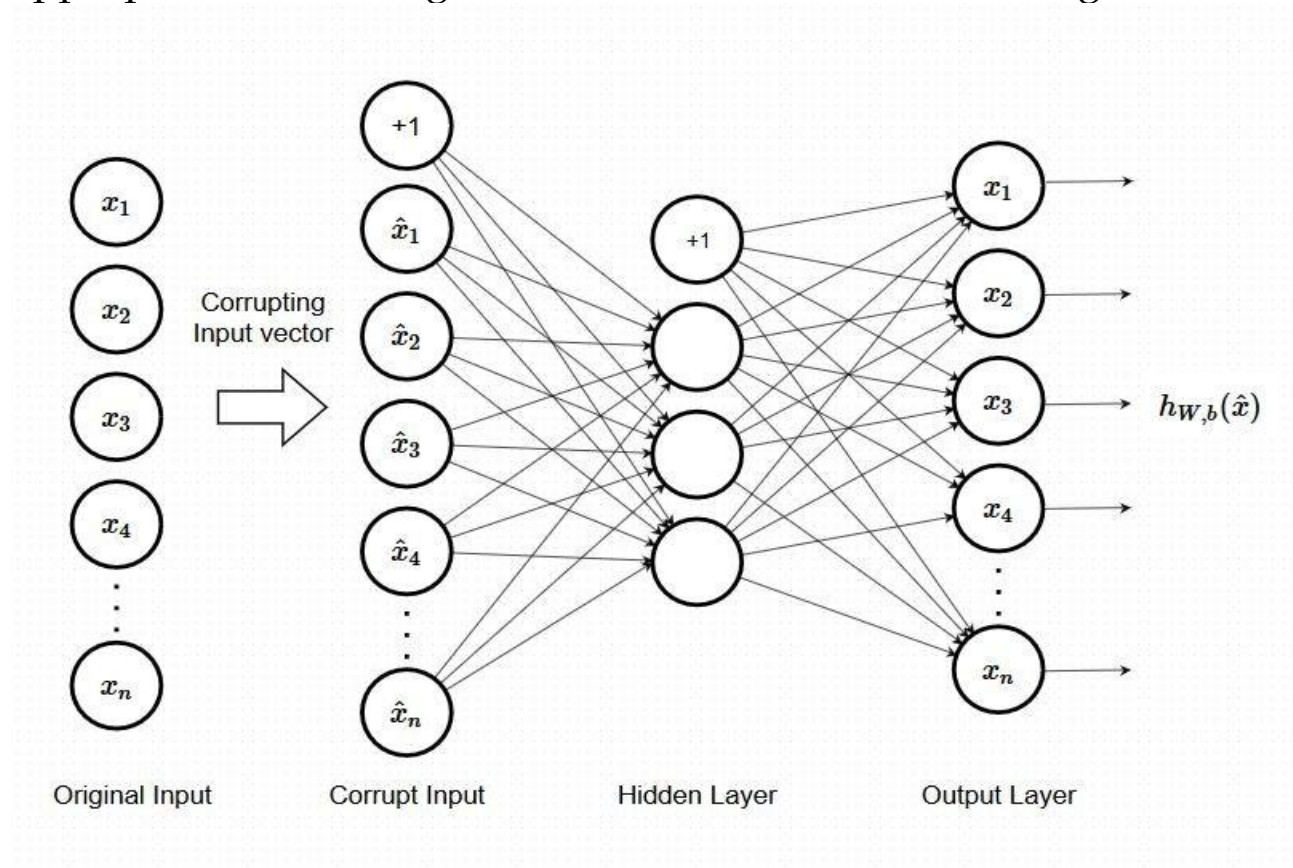
Input sequence x

$$\begin{matrix} x_1^t & \dots & x_1^0 \\ x_2^t & \dots & x_2^0 \\ x_3^t & \dots & x_3^0 \end{matrix}$$




- **Denoising AutoEncoder (DAE)**

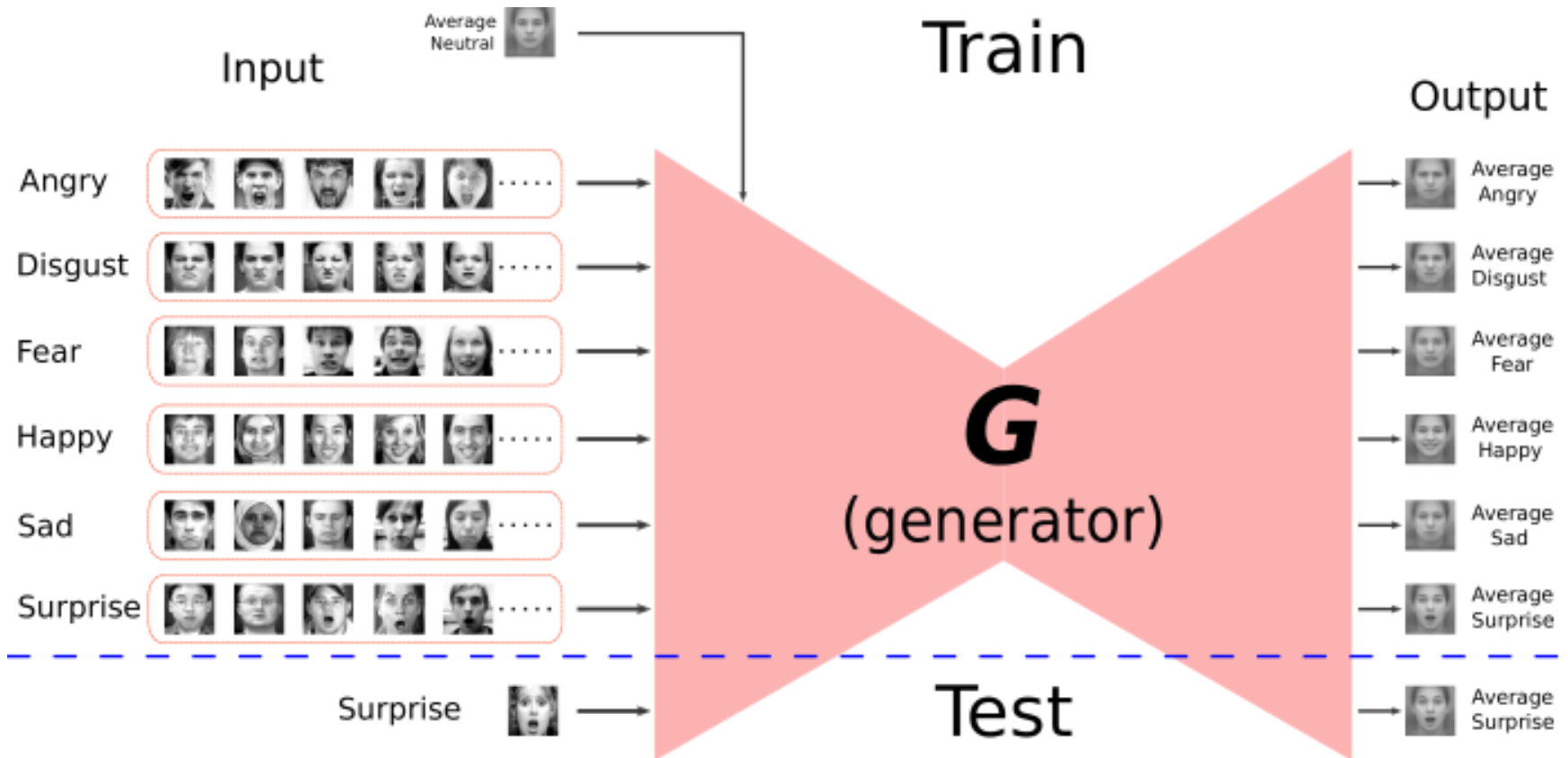
- Input is viewed as «corrupted»
- An appropriate set of weights is estimated to act as denoising filters



Networks of the previous scheme



- **Generative Adversarial Networks (GAN)**
 - See the deepfake chapter



The problem of «true» expressions



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Laboratory



V.S.



Real World



Anger

Disgust

Fear

Happiness

Neutral

Sadness

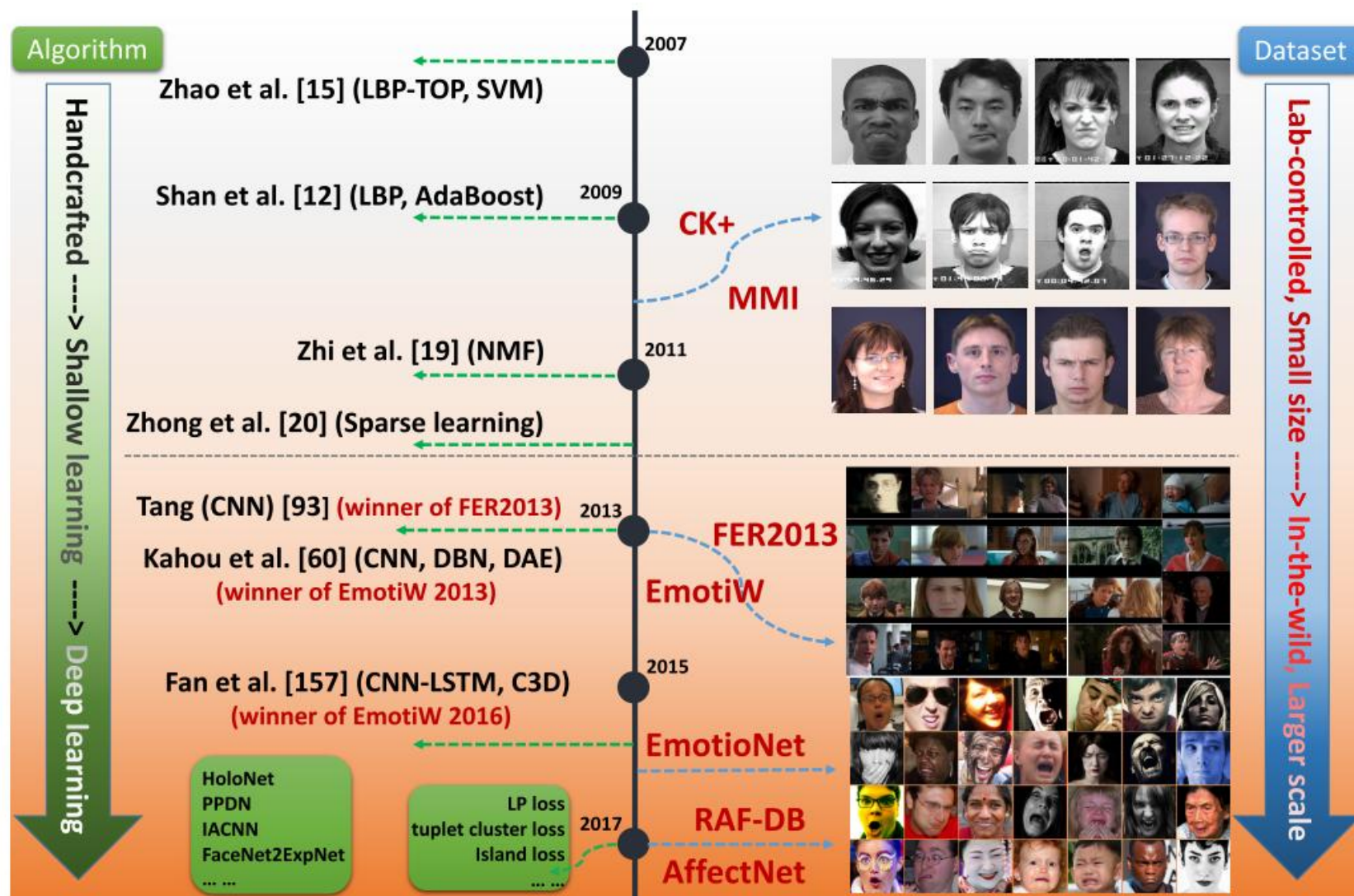
Surprise

Data sets and facial expression



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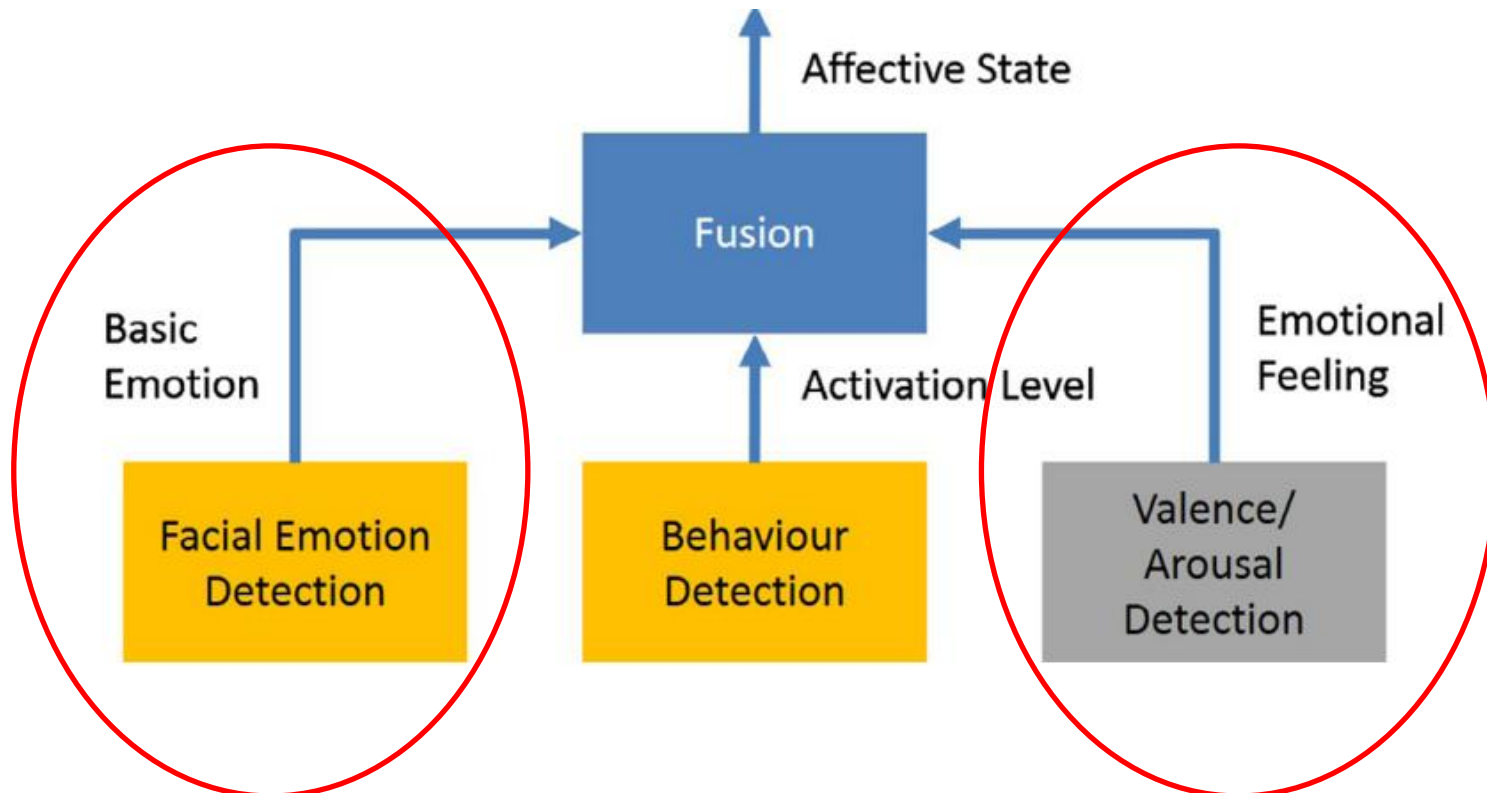
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Main issues

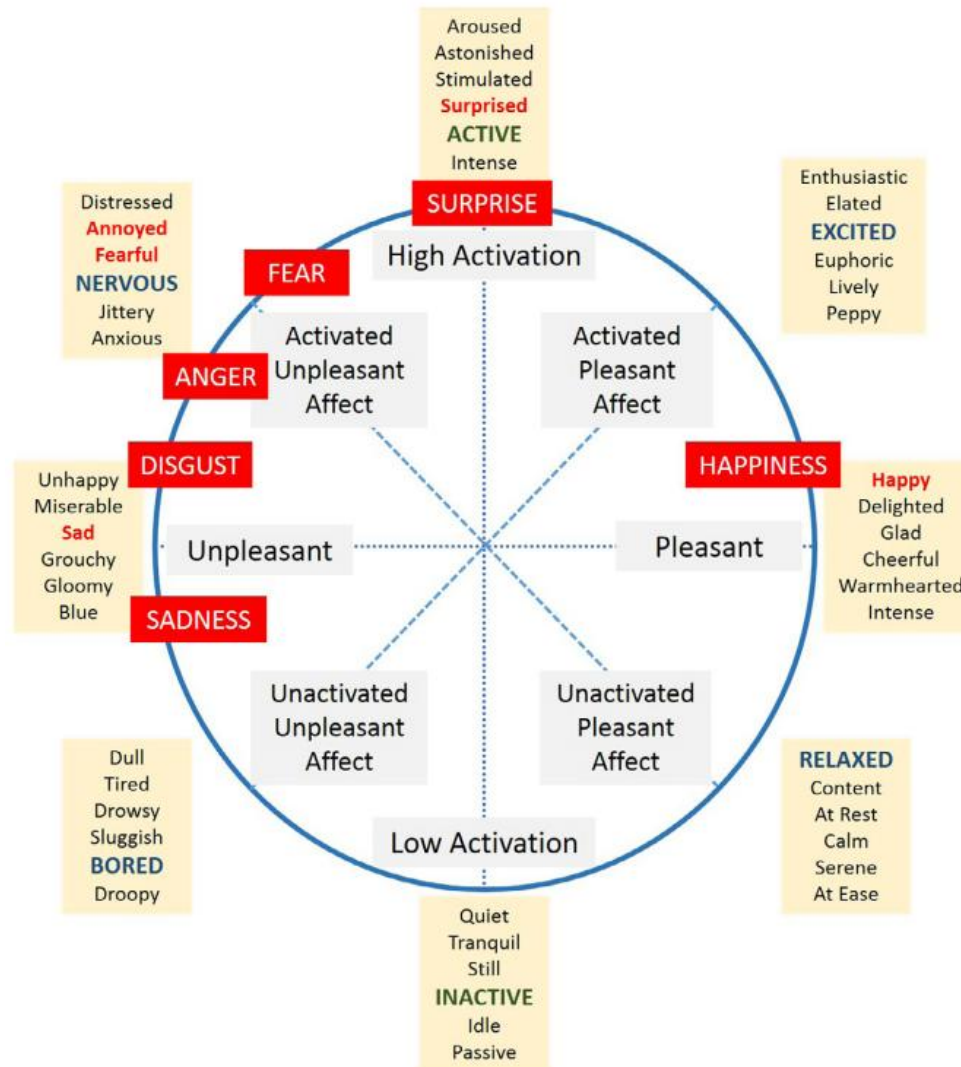


- Is it possible to find significant correlation between AUs and portions of the face «learnt» by DNs?
- How to verify if the accuracy is *explainable*?



- **Russell, 1980**
- **People can be viewed as «naive» psychologist**
 - «They have learned about emotions, organizing and summarizing their knowledge into a *cognitive structure*» (Russell, 1980)
 - They made the same «misunderstandings» of the machines in categorizing certain expressions/emotions (e.g. surprise, disgust and contempt)
- **This and further studies led to a graphical model where pleasantness-unpleasantness and sleep-arousal states are kept together**
 - Valence: pleasantness degree of the emotion
 - Arousal: activation degree of the emotion

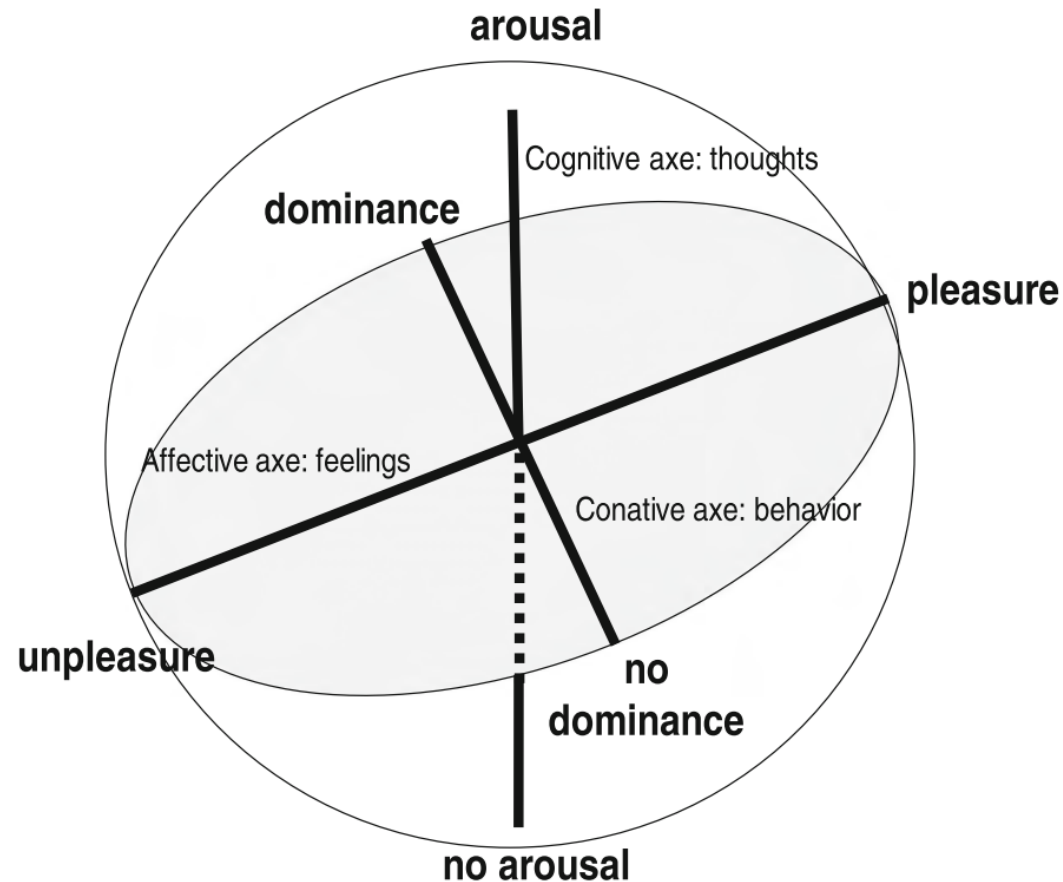
The circumplex model to emotions recognition



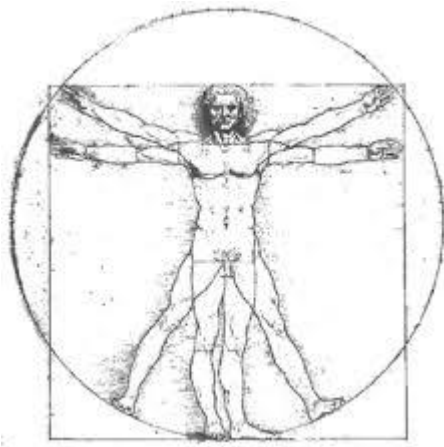
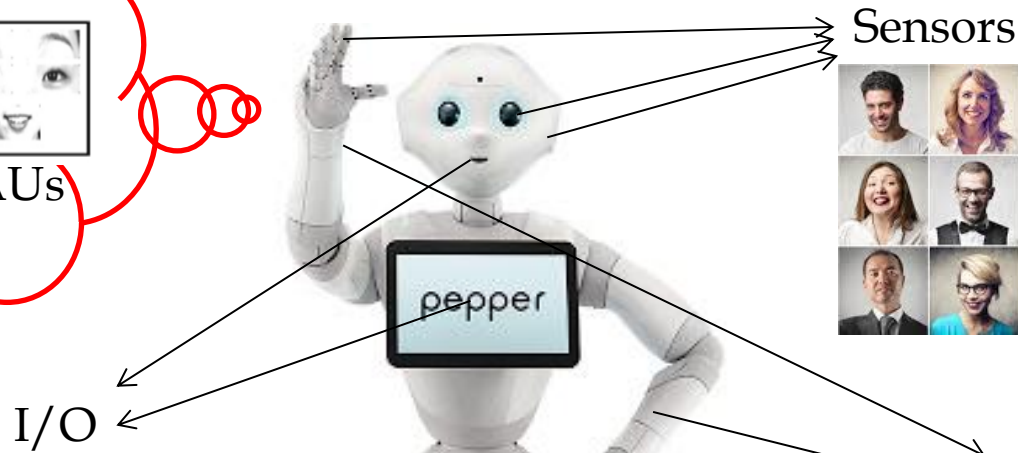
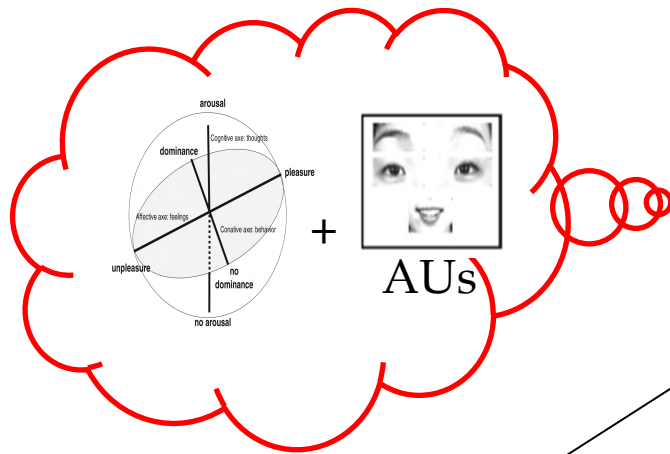
A more complete framework



- Bakker et al., 2014
- Dominance level is connected to feelings of control and behaviour restrictions caused by physical or social barriers
- In other words, to the **degree of control** that the person can apply to its feelings
- This can attenuate or increase the external perception of that emotion



Affective computing as novel paradigm



Actuators



«Understanding the human behaviour is the feedback for the machine to refine its answer»

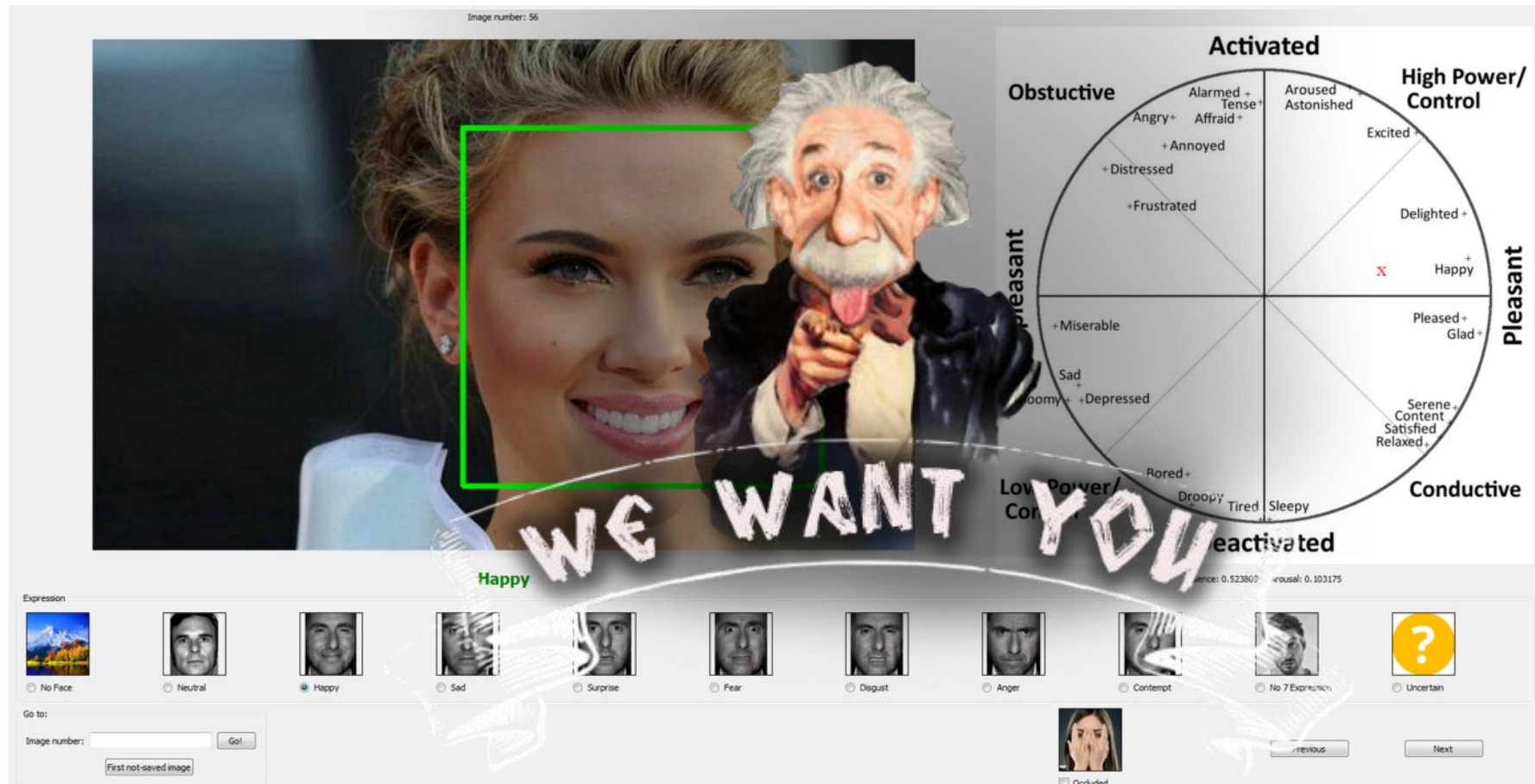
A. Turing, *The imitation game*

The challenge of AffectNet



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- First data set made up of categorical-based (AU) and continuous-based (VA) facial expressions



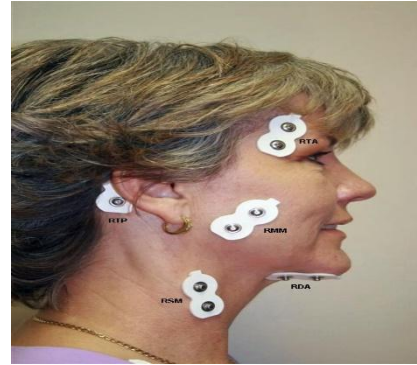
Other biometric traits for emotions/expressions recognition



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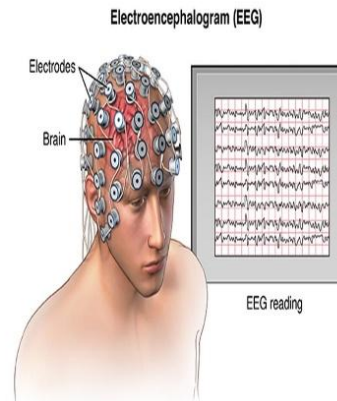
ElectroCardioGram
ECG



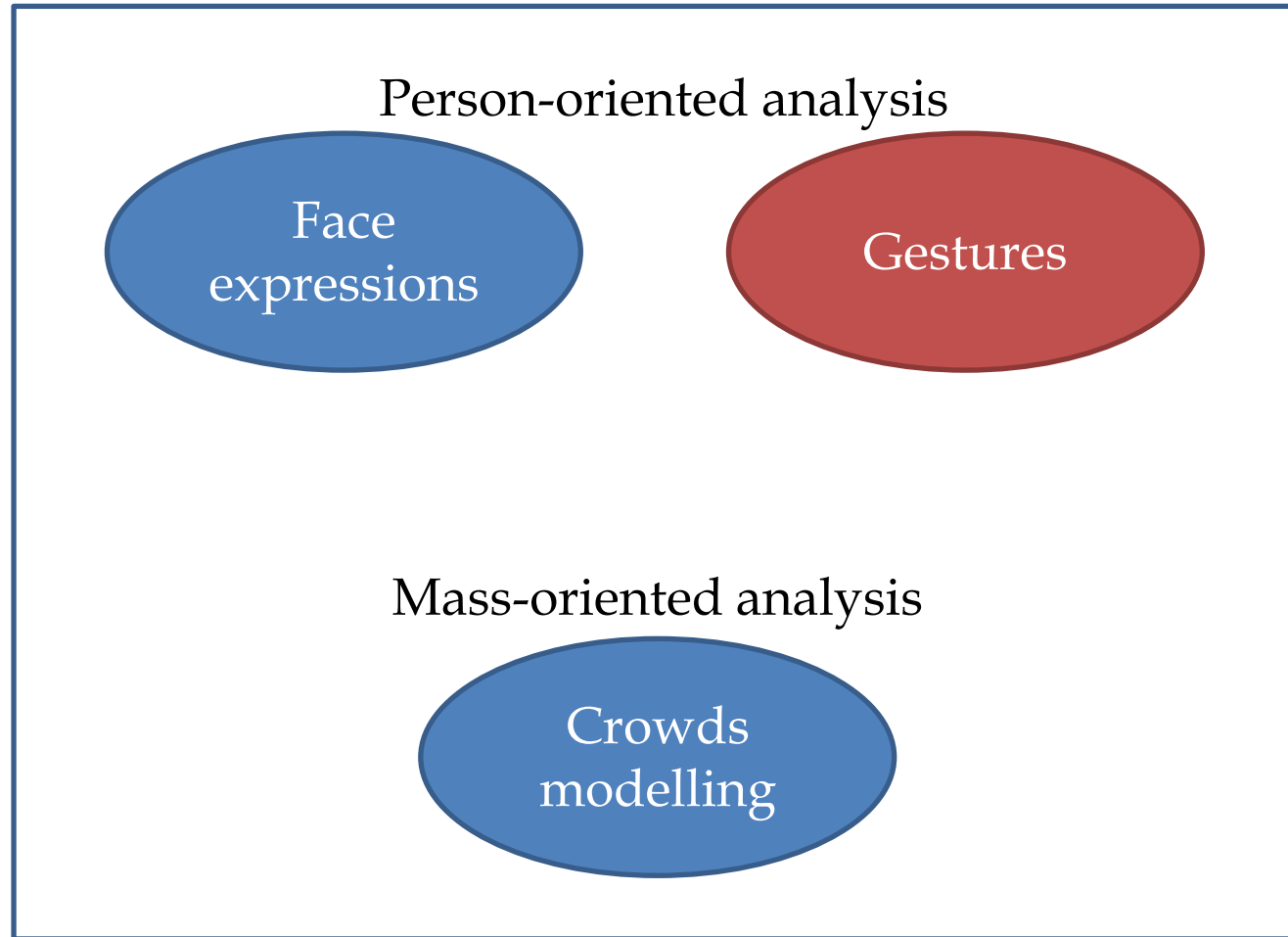
ElectroMyoGram
EMG



Thermogram



ElectroEncephaloGram
EEG

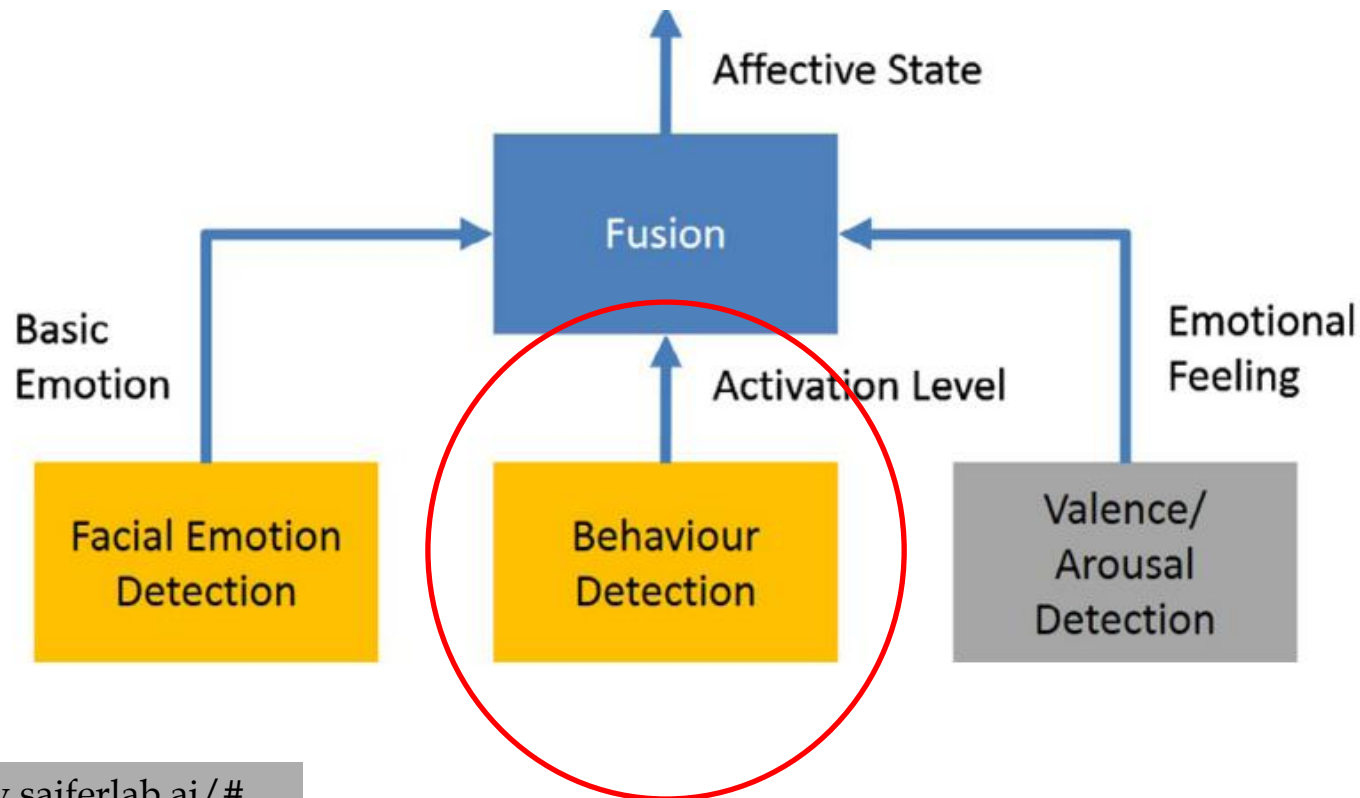


**Emotion and behaviour
recognition**

Gesture recognition



- **Person-level behaviour**
- **Applications**
 - Human-Machine Interface (interaction with bots/driving commands)
 - Interpreting the physiological/psychological state of a person for security



Two view-points



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Human-Machine interface



- Unnatural vs. natural gestures
- Commands vs. behaviour

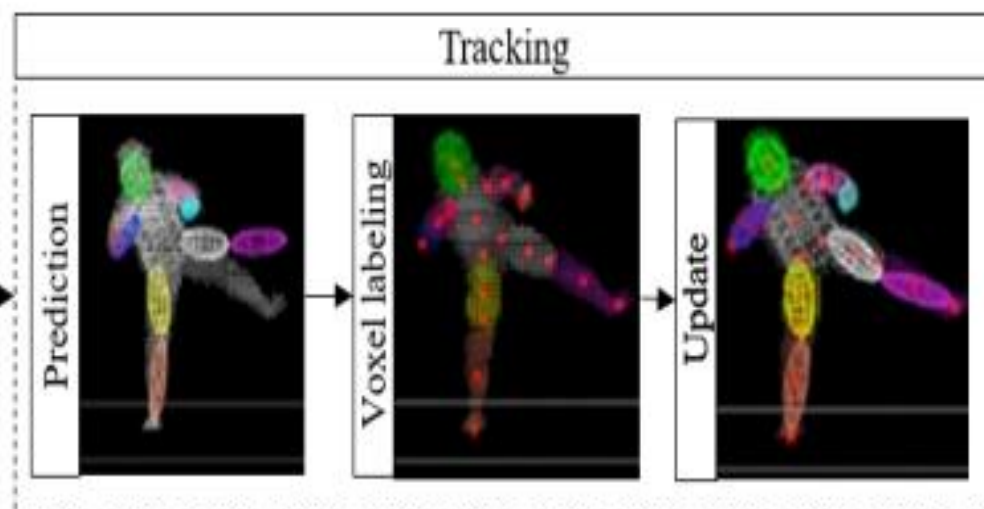
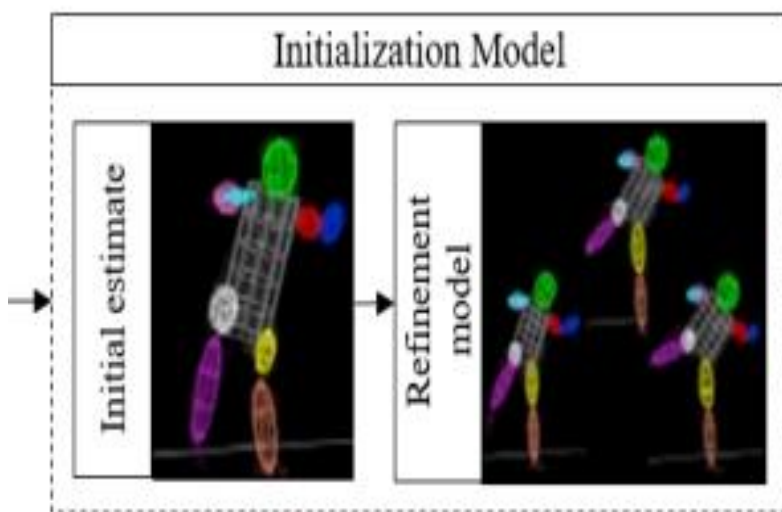
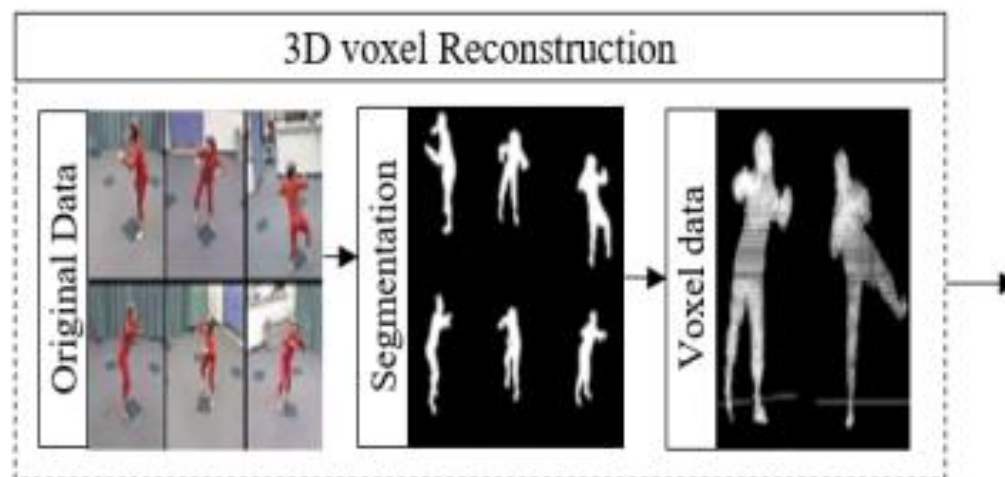
Human behaviour analysis



Basics



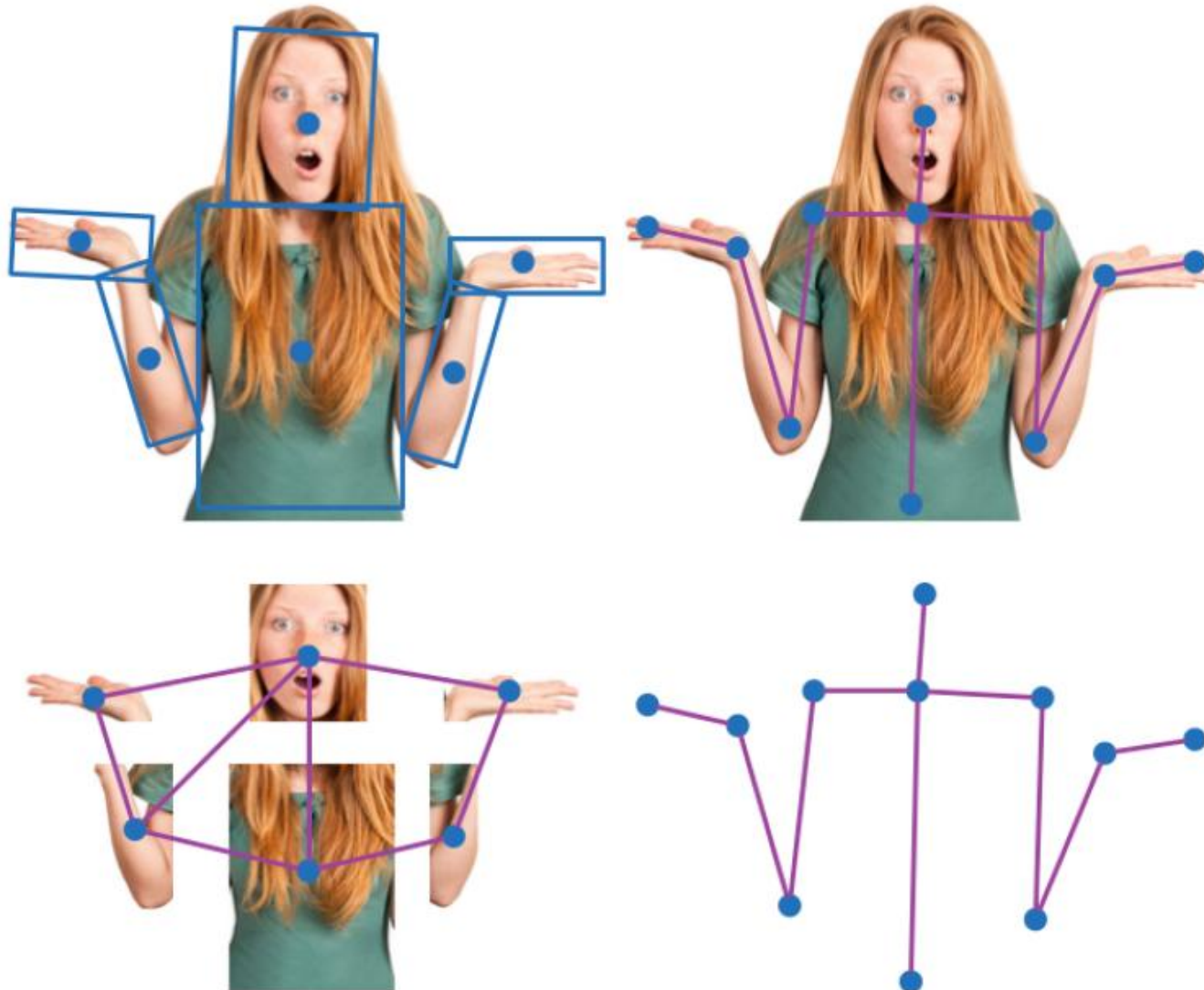
- Scene segmentation
- Body feature detection
- Body feature tracking



Using Kinect-like sensors



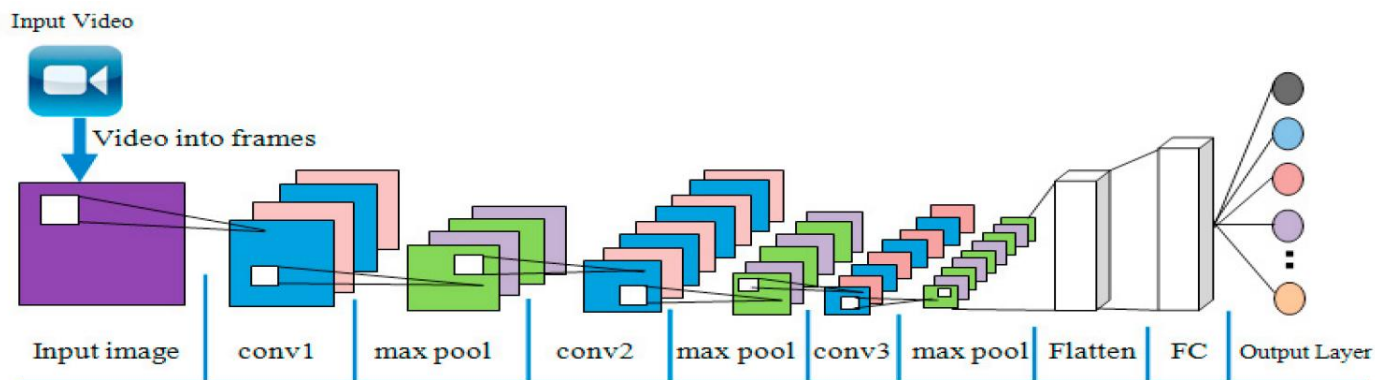
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CNN for human emotion recognition



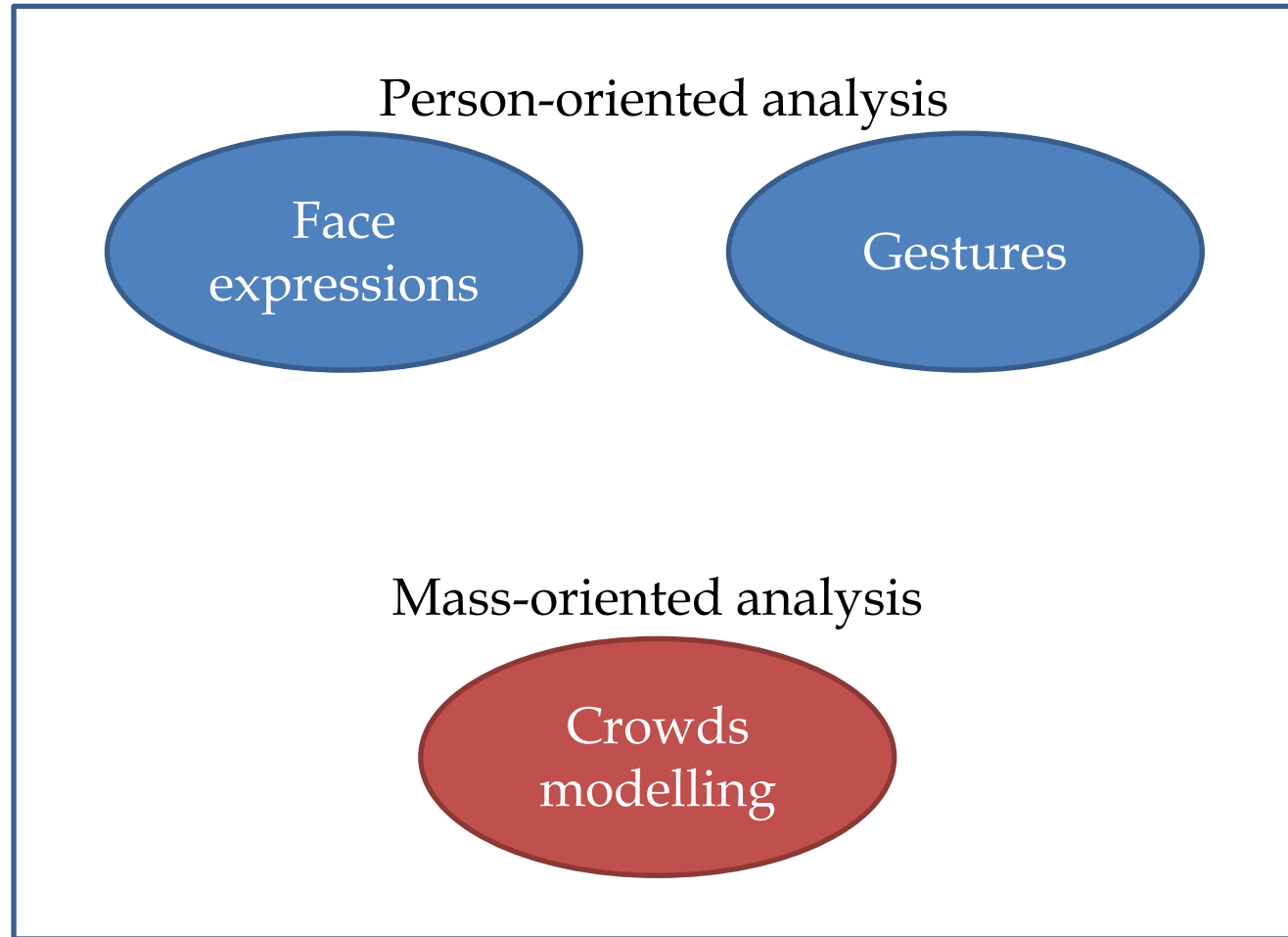
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Open issues



- **Detecting the natural behaviour is like detecting what people feel while talking**
- **Features availability**
- **Effective models to represent a «gesture»**
- **Realistic data sets in-the-wild**
 - With enough labelled information

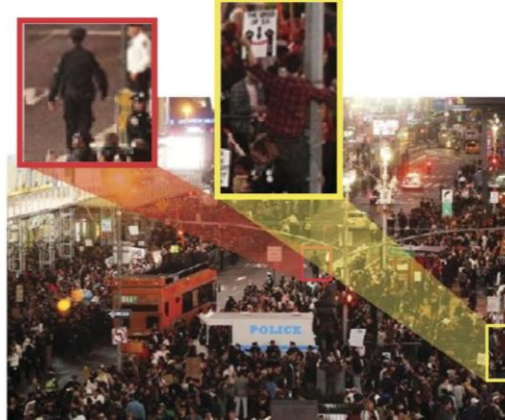


**Emotion and behaviour
recognition**

From person-based recognition to mass-based recognition

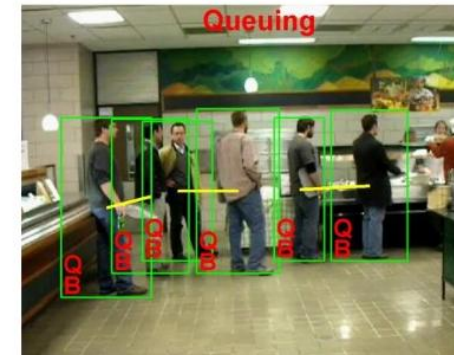
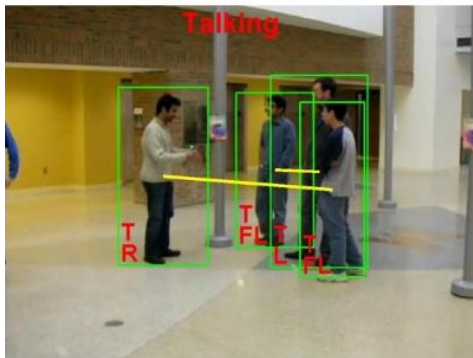


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CROWD ANALYSIS

ANOMALY DETECTION

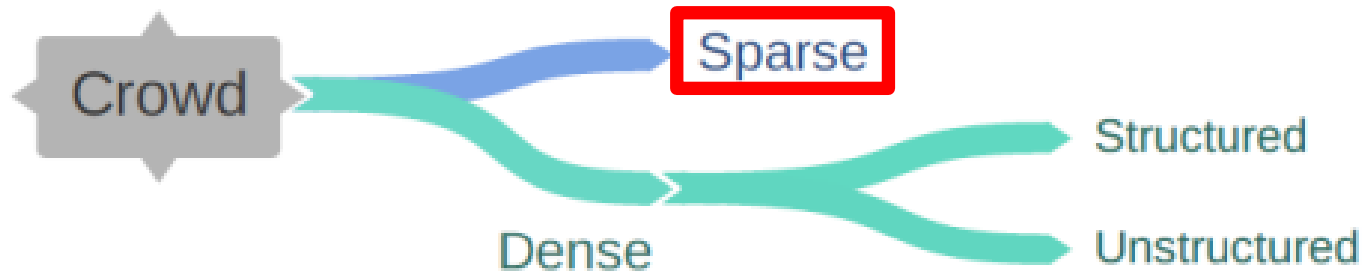


HUMAN BEHAVIOR RECOGNITION

Crowds definition



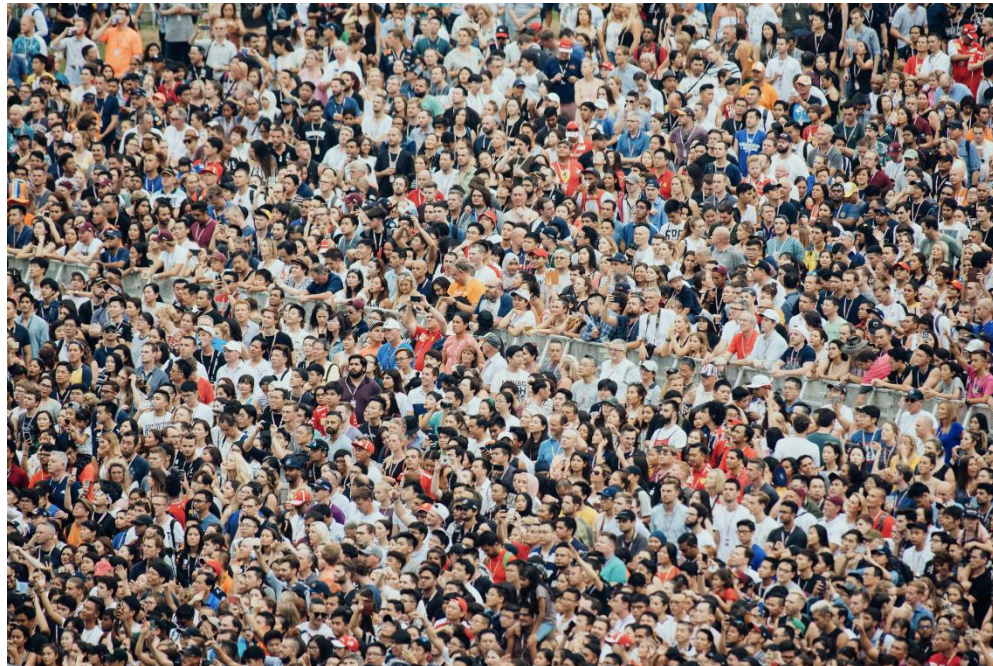
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Crowds definition



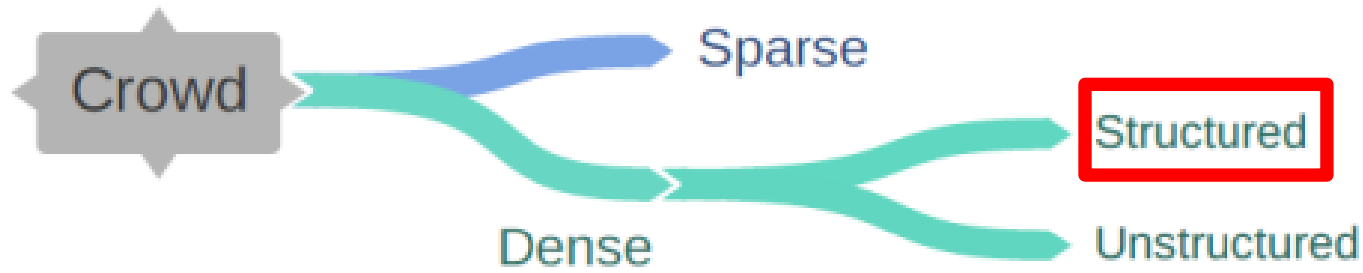
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Crowds definition



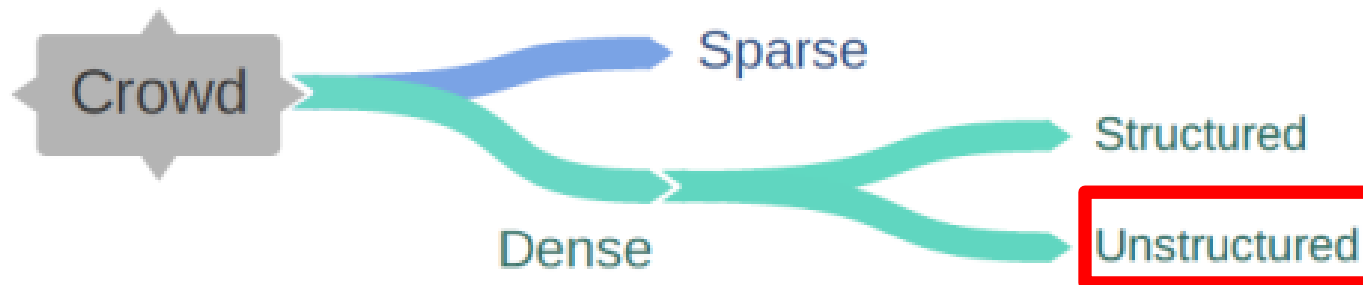
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Crowds definition



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Crowd density

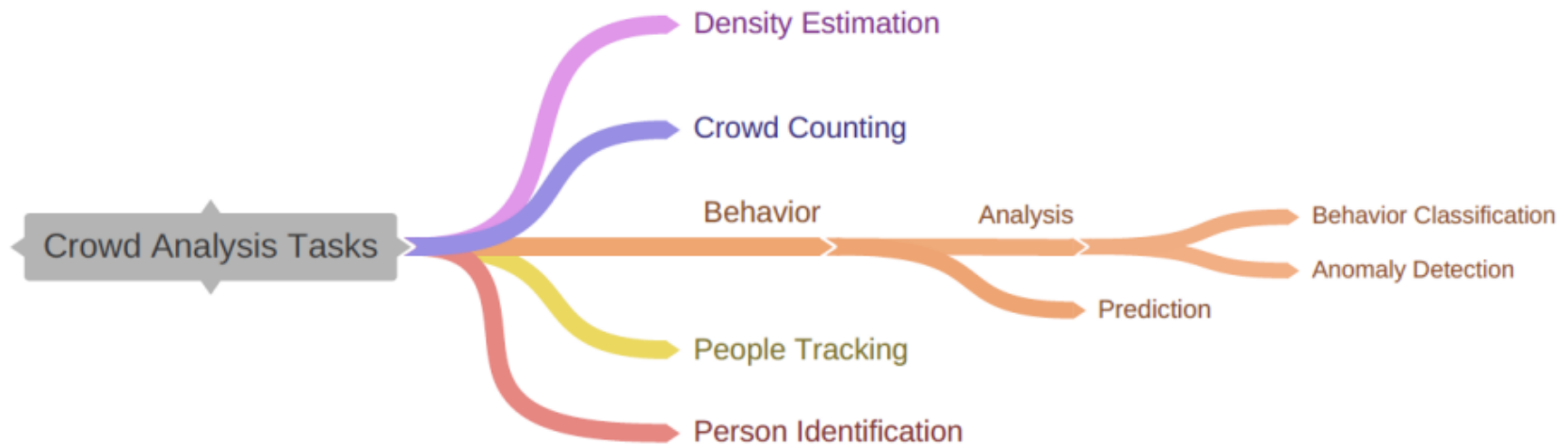


$$D = \frac{\#people}{\#area}$$

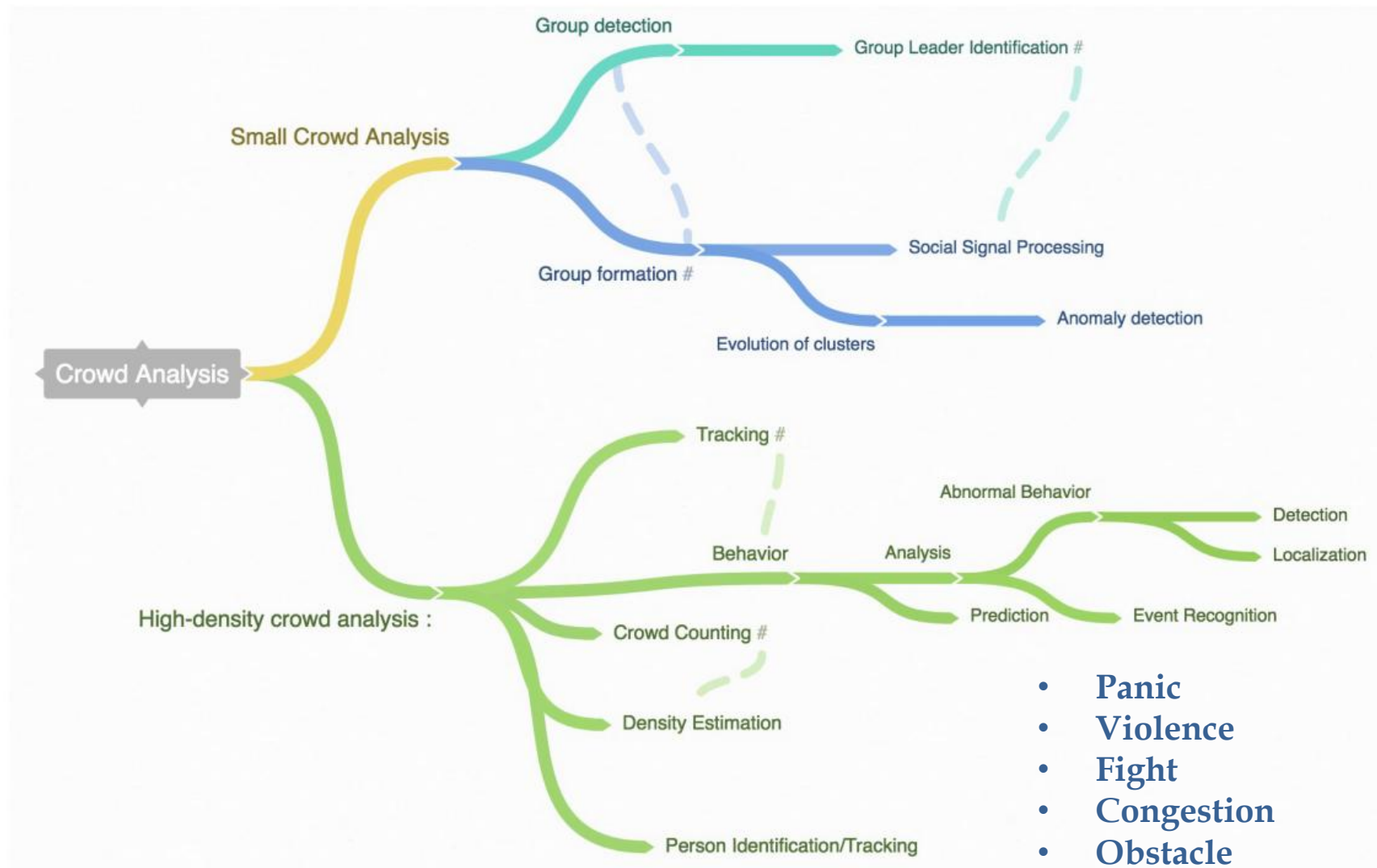
Measurement unit

1 mq can contain up
to 4 people

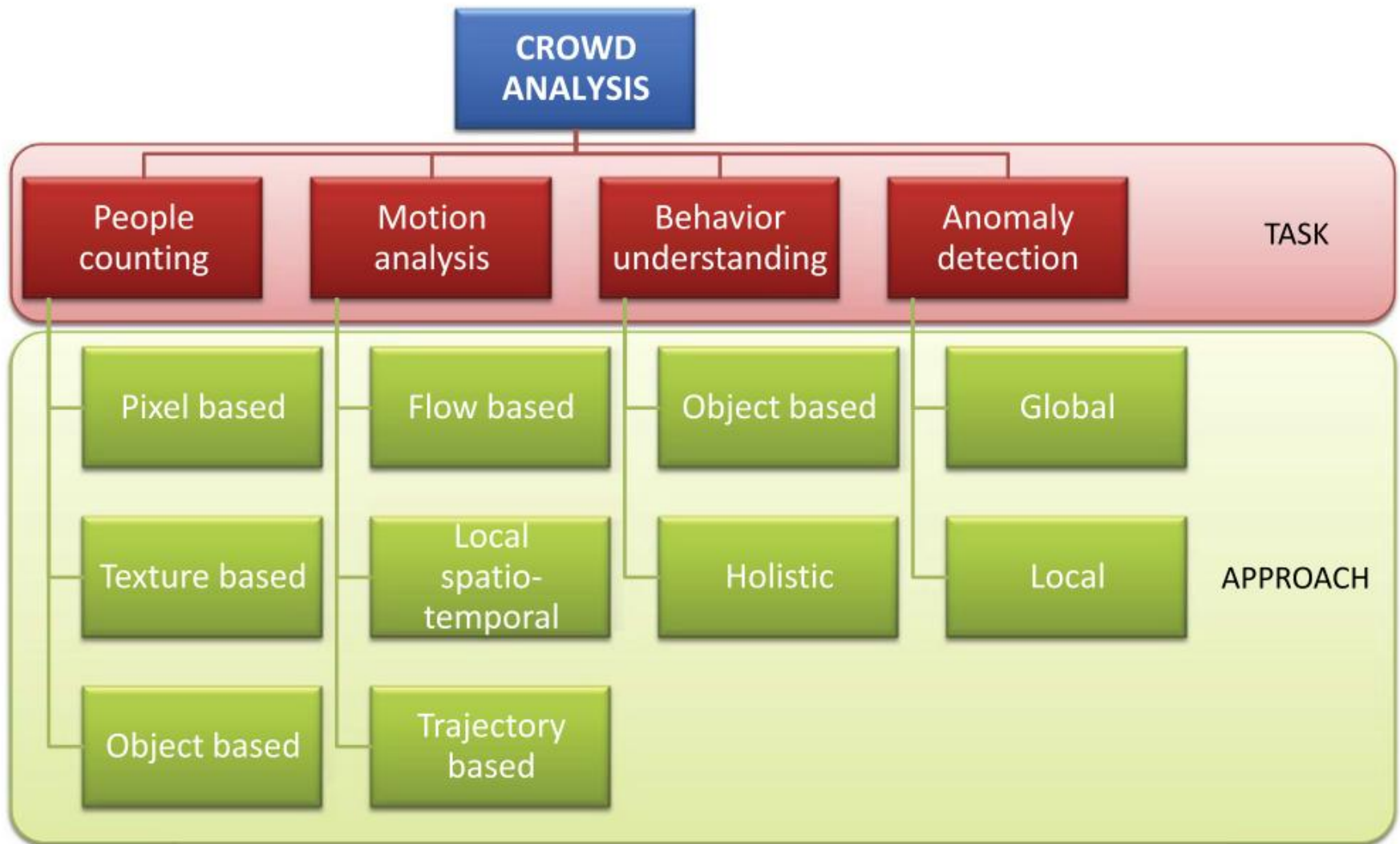
Crowd analysis



A deeper view



Detailed taxonomy

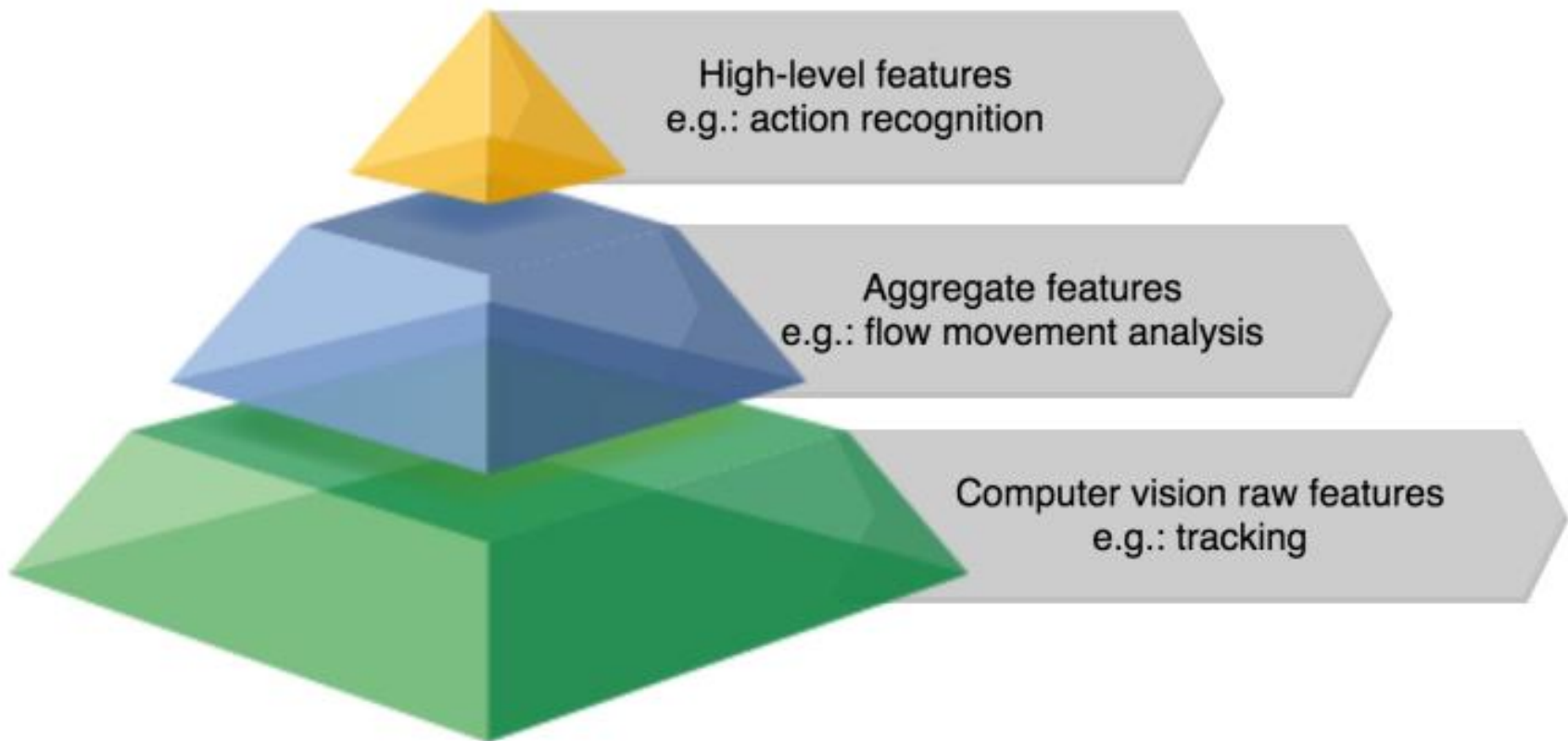


- Human behavior is extremely complex and exhibits large variations based on context and settings
 - Age, gender, body characteristics, cultural background and other factors.
 - No mathematical models including all possible cause-effect relationships
- The resolution of the visual sensor used for capturing the scene can strongly affect the analysis
- Crowds tend to be heterogeneous in nature and within the scene some part of the crowd can behave very differently



- **Definition: anomalous changes in the crowd density**
- **Social Force Models (SFM)**
 - represent the interaction force among individuals in a crowd using the repulsive and attractive forces
 - a model calibrated on a set of empirical observations may often fail to deal with a different set of observations
- **Flow Field Models (FFMs)**
 - capture crowd regular patterns and find how such patterns change over time
 - movements deviating from the regular patterns are considered as anomalies
- **Crowd Energy Models (CEMs)**
 - simulating how crowd motions transfer from normal to abnormal situation (and vice versa) using different concepts of physics energy such as potential energy, kinetic energy, entropy, and pressure

A bottom-up scheme



A hybrid Social Force-Flow Field model



SLOW DISGREGATION



SLOW AGGREGATION



FAST DISGREGATION



FAST AGGREGATION

Crowd transitions/states



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States

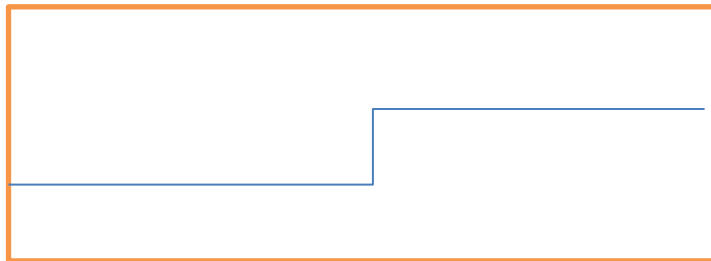
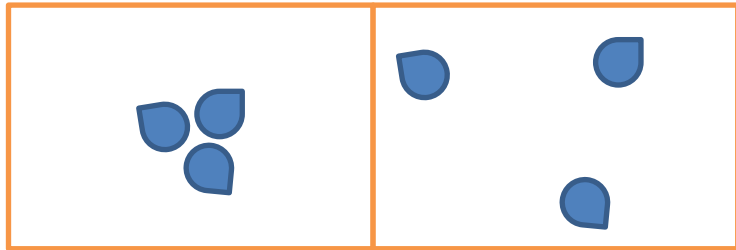
Normal/Abnormal

Actions

Aggregation/Disgregation

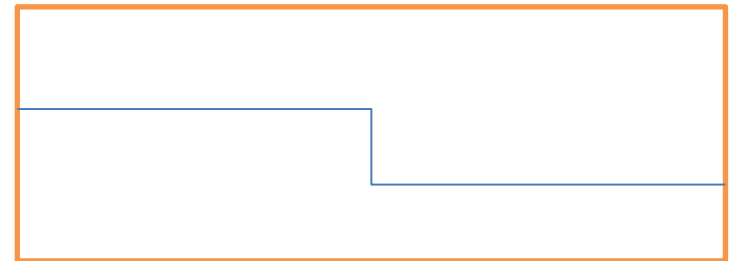
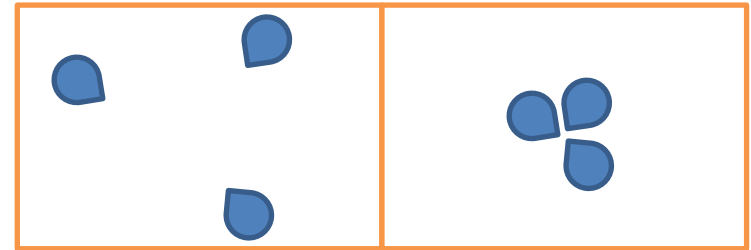
Modality

Slow/Fast (Anomaly)



DISGREGATION

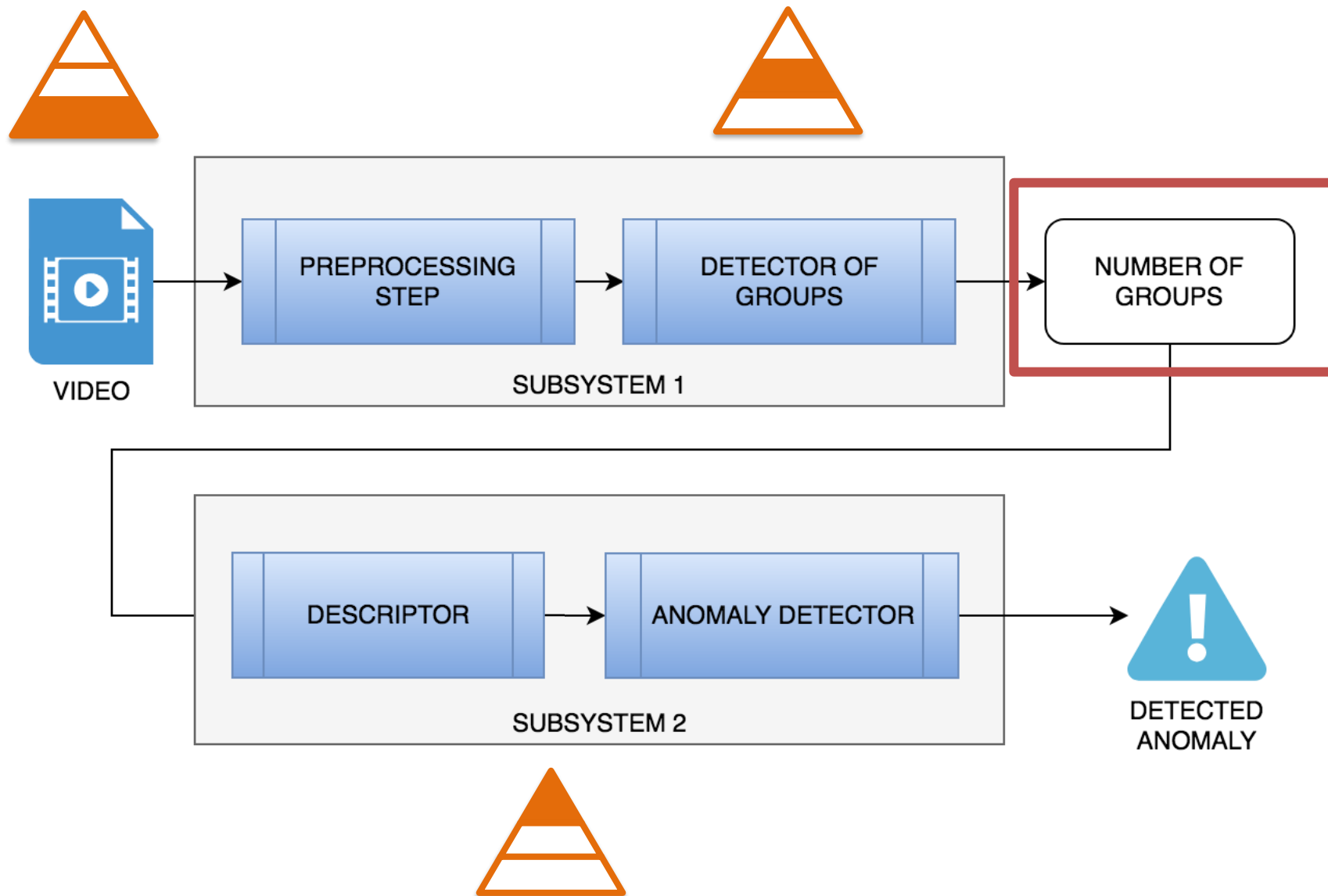
TRANSITION 01



AGGREGATION

TRANSITION 10

System modules



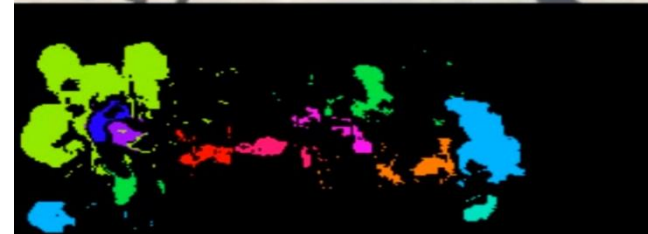
Group detection and counting



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MANUAL COUNTING



OPTICAL FLOW

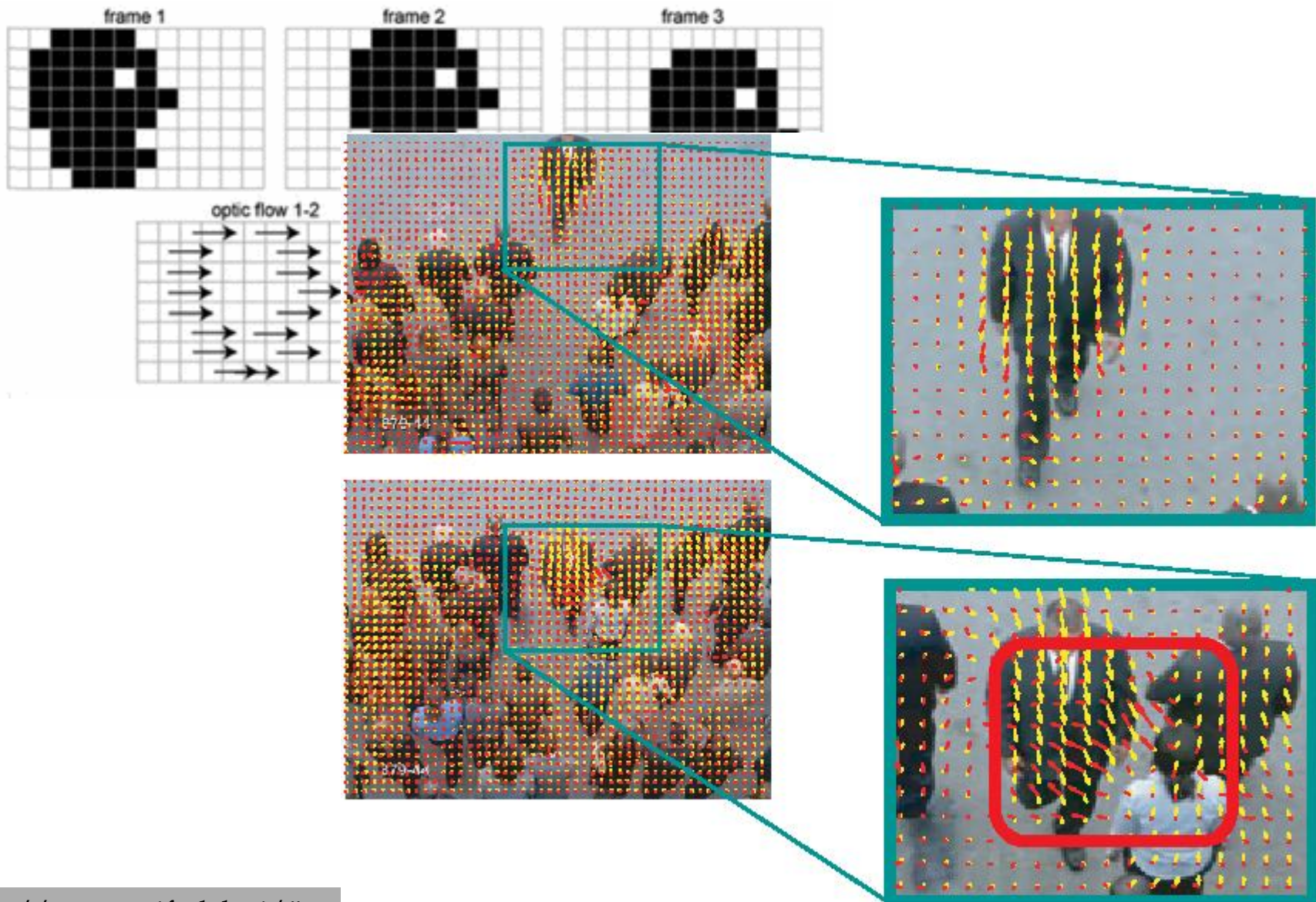


PERSON DETECTOR



BLOB DETECTOR

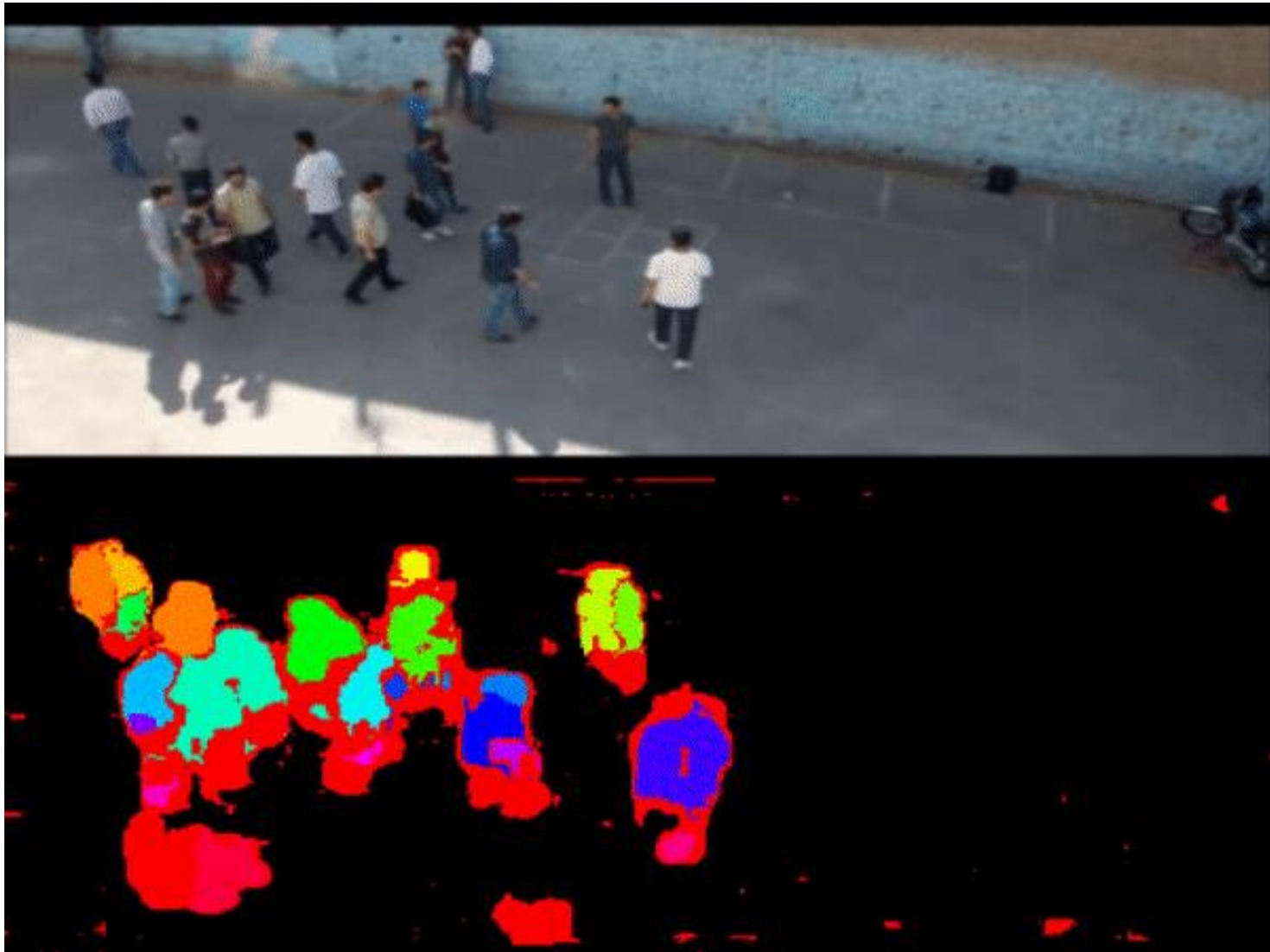
Optical flow for anomaly detection



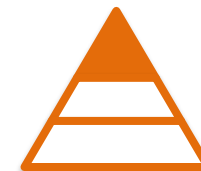
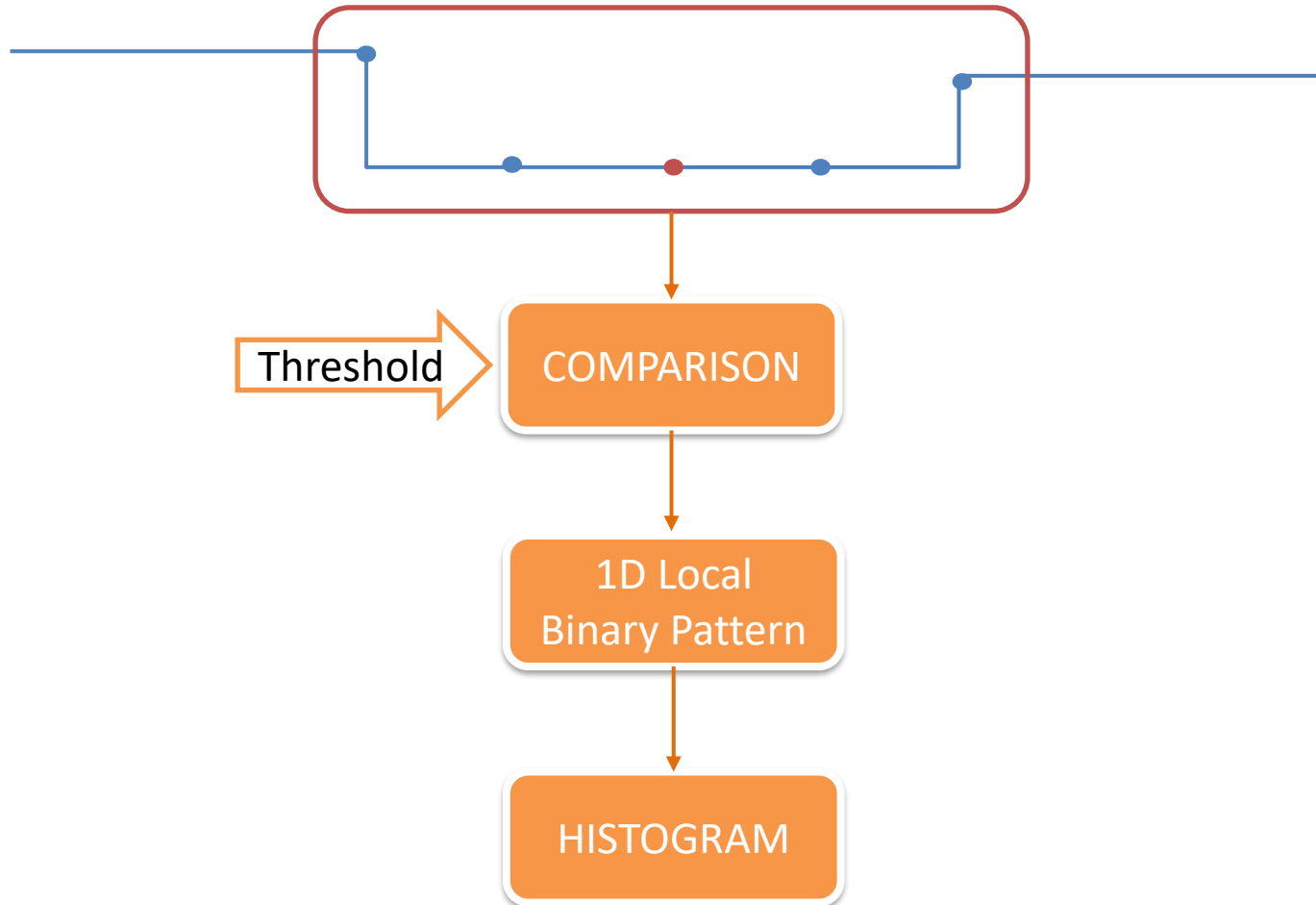
Group detection by optical flow



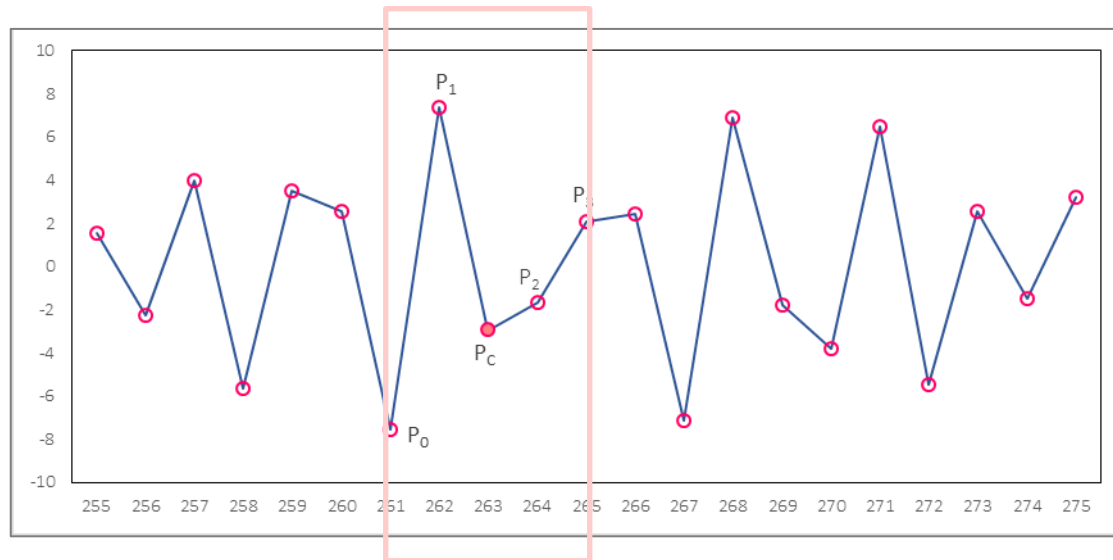
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Crowd dynamic descriptor



1D Local Binary Pattern



-7,5	7,4	-3,0	-1,7	2,1
------	-----	------	------	-----

0	1		1	1
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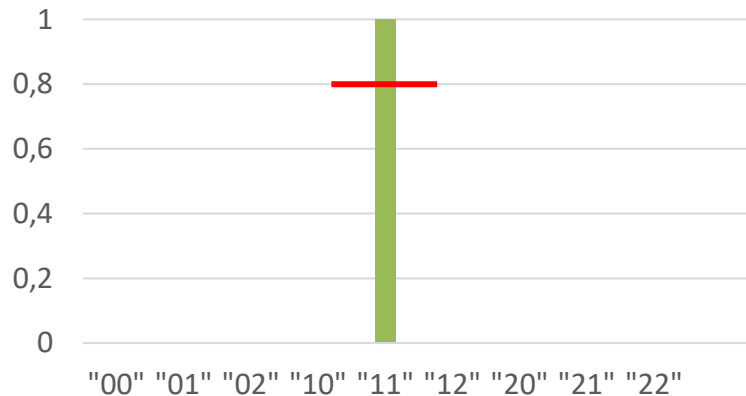
7

Abnormal event detection



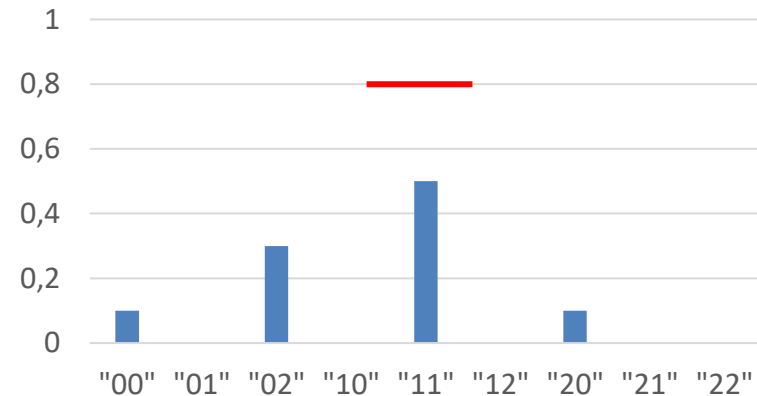
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NO ANOMALY

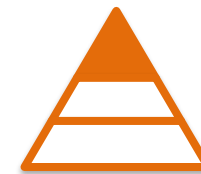


Central bin → Normal

ANOMALY



Other bins → Abnormal



Evaluation parameters



- **False Positive/False Negative (system's disagreement)**
 - Missclassified normal events (FP)
 - Missclassified abnormal events (FN)
- **True Positive/True Negative**
 - Correctly classified abnormal events (TP)
 - Correctly classified normal events (TN)

- **False Positive Rate/False Negative Rate**

$$FPR = \frac{FP}{FP + TP}$$

$$FNR = \frac{FN}{TN + FN}$$

- **Precision/Recall**

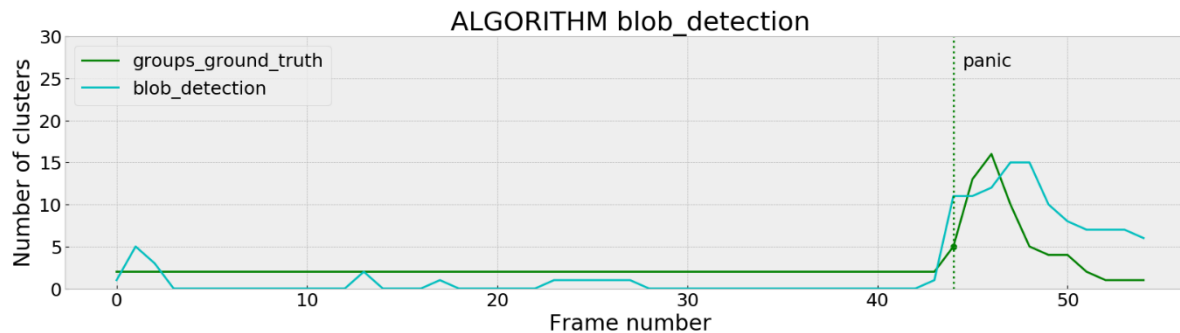
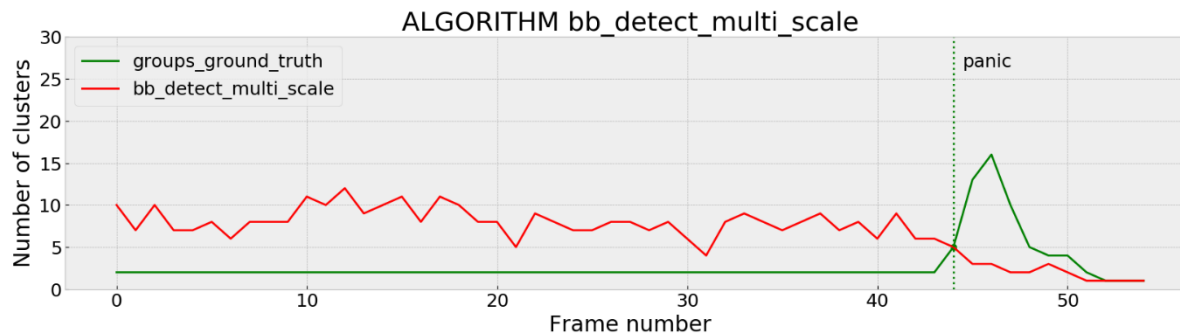
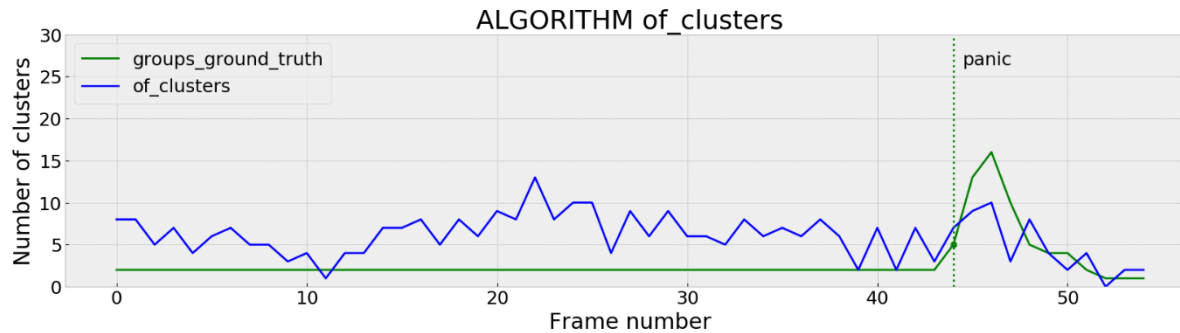
$$Precision = \frac{TP}{TP + FP}$$

Human's
viewpoint

$$Recall = \frac{TP}{TP + FN}$$

System's
viewpoint

Group counting – some examples

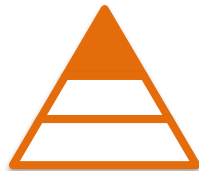
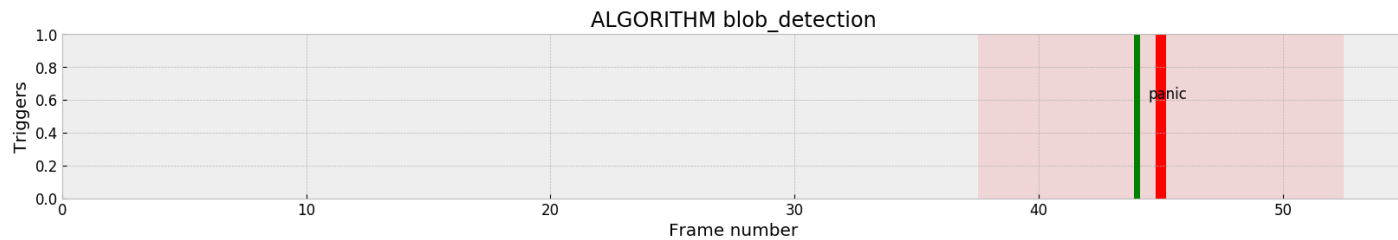
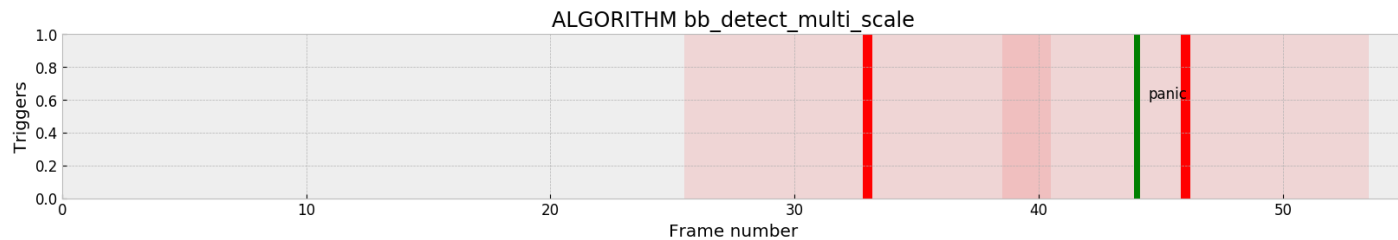
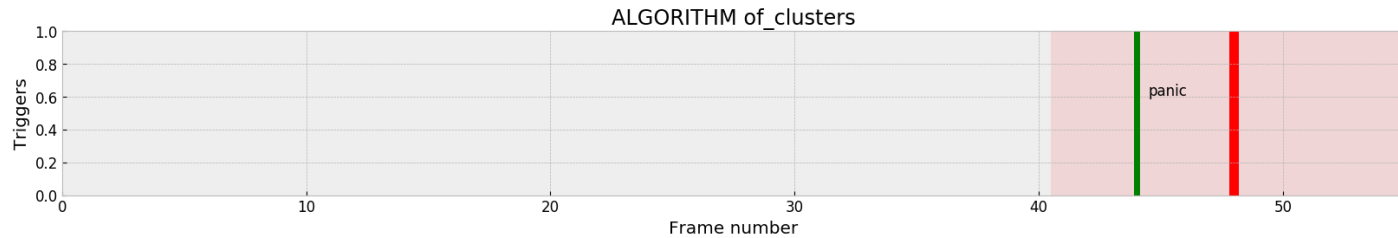
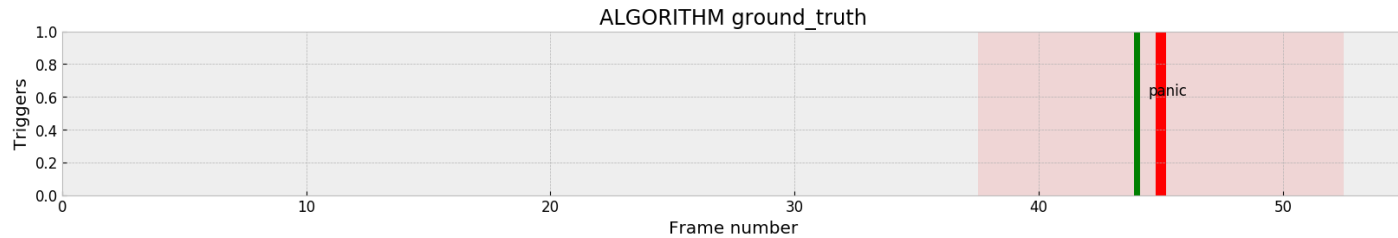


Anomaly detection – some examples



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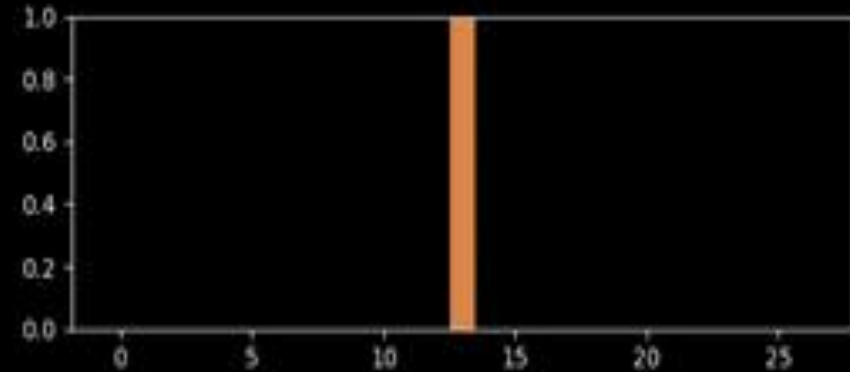
VIDEO 009



Violence detection



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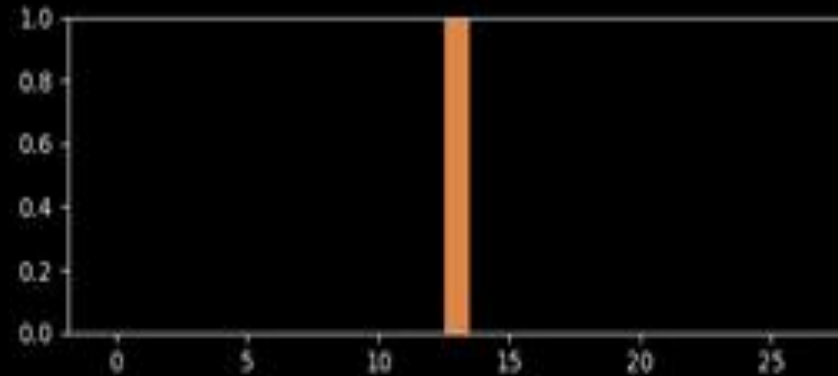


G. Orrù, D. Ghiani, M. Pintor, G.L. Marcialis, F. Roli, Detecting Anomalies from Video-Sequences: a Novel Descriptor, IEEE/IAPR 25th Int. Conf. on Pattern Recognition (ICPR 2021), Milano (Italy), 10-15th, Jan., 2021, <https://arxiv.org/abs/2010.06407>, DOI: 10.1109/ICPR48806.2021.9412855.

Panic detection



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G. Orrù, D. Ghiani, M. Pintor, G.L. Marcialis, F. Roli, Detecting Anomalies from Video-Sequences: a Novel Descriptor, IEEE/IAPR 25th Int. Conf. on Pattern Recognition (ICPR 2021), Milano (Italy), 10-15th, Jan., 2021, <https://arxiv.org/abs/2010.06407>, DOI: 10.1109/ICPR48806.2021.9412855.

Open issues

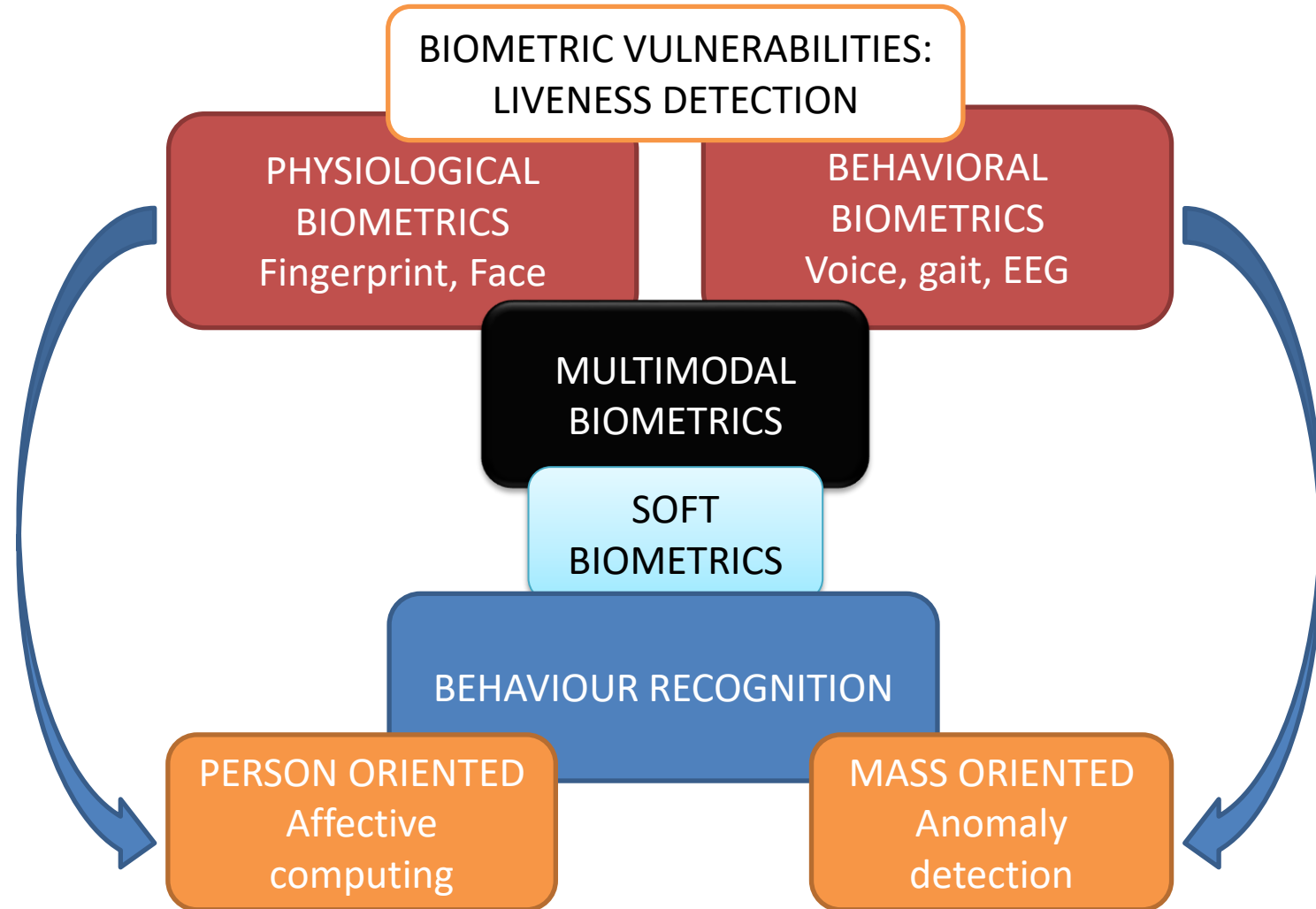


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- Definition of «crowd»
- General purpose methods?
 - Camera/action viewpoints
- Computational complexity
- Optimal descriptor
- Data
- How about deep learning?



Concluding remarks





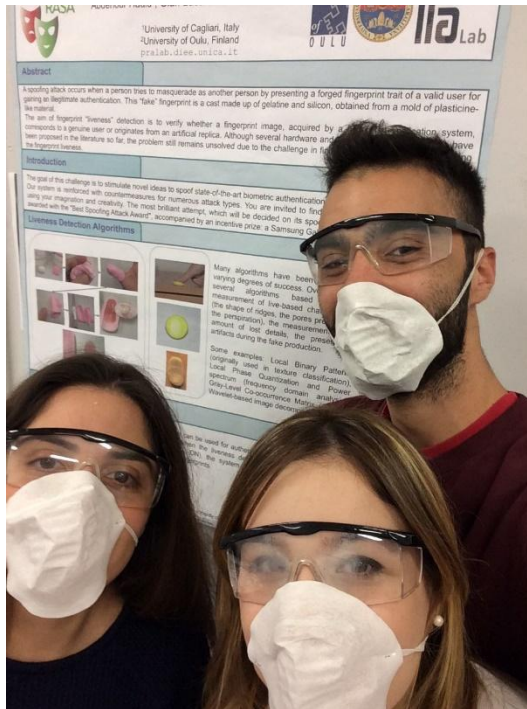
That's all!

Now it's
your
turn!





Thank you for listening!



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